MILITARY SPECIFICATION

PERISCOPES, TANK: XM44E1 AND XM44E2

1. SCOPE

1.1 This specification covers the performance requirements, related quality assurance provisions, and packaging data for the Periscopes: XM144E1 and XM44E2. The periscopes provide a unity-power viewing system and a nine power passive (low light level) sighting system. The periscopes are equipped with a reticle projector which projects the image of a reticle pattern into the sighting system.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Material: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Material, Packaging and Packing of
MIL-O-14850	Oscillator, High voltage: 10549785
MIL-I-45607	Inspection Equipment, Acquisition and Disposition of
MIL-I-55536	Image Intensifier Tube Assembly, 25MM

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-150	Photographic Lenses
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-810	Environmental Test Methods

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DRAWINGS

US Army, Frankford Arsenal

F10542200	Periscope, Tank: XM44El
F11731100	Periscope, Tank: XM44E2
F8565556	Pressure Tester-Two Station
F11738178	Gage - Interchangeability - Maximum
F11738179	Gage - Interchangeability - Minimum
F11738184	Fixture - Interchangeability - Pressure Testing

PACKAGING DATA SHEETS

10542200	Packaging of Periscope: XM44E1
11731100	Packaging of Periscope: XM44E2

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection With specific procurement functions should be Obtained from the procuring activity Or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The periscope shall be manufactured in accordance with the applicable drawing pertaining thereto (see 2.1).

3.2 Image intensifier tube. - The image intensifier tube shall conform to MIL-I-55536 (see 3.12.1).

3.3 High voltage oscillator. - The high voltage oscillator shall conform to MIL-O-14850.

3.4 Temperature conditions. - Except as specified in 3.7 and 3.8 the following requirements shall be met at a temperature between +60°F and +90 °F.

3.5 General specification.- The requirements of MIL-F-13926 listed below shall be met wherever applicable.

a. Order of precedence.
b. Dimensions and tolerances.
c. Inorganic protective surface finishes.
d. Part identification and marking.
e. Soldering electrical connections.

3.6 Workmanship. - The workmanship requirements of MIL-F-13926 shall be met. The exterior protective surface finishes shall be free of voids which may affect their protective qualities or appearance. Optical surfaces shall be free of scratches and blemishes which may affect their optical qualities or appearance.

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3.7 Extreme storage temperatures. - There shall be no evidence of physical failure such as glass breakage and bonding or casting failure due to storage at temperatures ranging from -80°F to +160°F.

3.8 Extreme operating temperature. - The periscope shall meet the applicable requirements of 3.13.1 and 3.13.2 while exposed to and thermally stabilized at -65°F and +150°F.

3.9 Vibration. - This requirement shall be met with:

a. The periscope assembled to its mount assembly.
b. The image tube, through the high voltage oscillator, and the reticle projector lamp energized using 6 volts dc.
c. A line of sight established prior to vibration.
d. The clear filter positioned in the optical train.

The periscope shall be vibrated successively along three (3) mutually perpendicular axes. Vibration shall be a simple harmonic motion, the frequency varying either linearly or logarithmically from 5 to 55 to 5 CPS in five minute cycles for a total of 15 minutes along each of the axes. The amplitude of vibration shall be maintained constant at 1/64 inch (1/32 inch total excursion). The vibration test shall not cause:

e. Failure or flicker of light emission from the reticle projector lamp, and image tube during vibration.
f. Evidence of physical failure during and subsequent to vibration.
g. The established line of sight to deviate more than 0.30 mil subsequent to vibration.

3.10 Dielectric strength. - This requirement shall be met with the image tube and the reticle projector lamp removed from their receptacles, Insulation resistance between ungrounded circuits, isolated circuits, and castings shall be not less than 10 megohms at a potential of 100 Vdc.

3.11 Optical characteristics, with image tube removed. - Unless otherwise specified the requirements 3.11.1 through 3.11.6 inclusive shall be met with:

a. The image tube removed from the periscope.
b. The periscope assembly to its mount assembly.
c. The clear filter positioned in the optical train.

3.11.1 Contrast transfer. - The contrast transfer for vertical lines and for horizontal lines shall be not less than the values specified in 3.11.1.1 and 3.11.1.2.

3.11.1.1 Objective system. - The following values (for the axis specified) shall be met with 2870° ± 100° Kelvin (K) radiance, modulated at the spatial frequency (angular spacing) indicated, incident to the entrance window and the contrast values obtained at the objective system focal plane. On axis is defined as the geometric axis of Sleeve 10542226.

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Off axis is defined as displacement, conforming to the value specified, from on axis.

ANGULAR SPACING in line pairs/milliradian)	CONTRAST TRANSFER Vertical lines	Horizontal lines
a. On axis.		
0.58	.95	.95
1.732	.91	.91
3.70	.84	.84
b. Off axis 0.3 inch.		
0.58	.93	.93
1.732	.88	.88
3.70	.78	.78

3.11.1.2 Eyepiece and erecting system. - The following values (for the axis specified) shall be met with 2870° ±100 Kelvin (K) - radiance, modulated at the special frequency (angular spacing) indicated, incident to the exit pupil of the eyepiece and erecting system and the contrast transfer values obtained at the focal plane.

ANGULAR SPACING in line pairs/milliradian)	CONTRAST TRANSFER Vertical lines	Horizontal lines
a. On axis.		
0.58	.95	.95
1.732	.92	.92
3.70	.81	.81
b. Off axis 0.3 inch.		
0.58	.95	.95
1.732	.92	.92
3.70	.77	.77

3.11.2 Luminous transmittance. - The requirements of 3.11.2.1 and 3.11.2.2 shall be met with a source of radiance at the focal plane of the system specified.

3.11.2.1 Objective system.

a. Luminous transmittance. - Liminous transmittance shall be not less than 58% when observed through any area of the entrance pupil of the periscope.

b. Transmittance at 0.85 micron. - Transmittance at 0.85 micron shall be not less than 35%.

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3.11.2.2 Eyepiece and erecting system. - This requirement shall be met when observing through the periscope exit pupil. Luminous transmittance shall be not less than 6077.

3.11.3 Head assembly. - The following requirements shall be met with the body assembly removed from its mount.

a. Luminous transmittance. - Luminous transmittance shall be not less than 90%. The source of radiance shall be at the exit aperture of the head assembly and the observation made at the entrance aperture.

b. Resolution. - Resolution shall be 2.0 seconds of arc or less when observing with a telescope of at least 90 power. The entrance pupil of the telescope shall be not less than 7.0 inches. Resolution shall be as observed at one focus setting of the telescope.

3.11.4 Resolution - periscope objective system. - The requirements of 3.11.4.1 and 3.11.4.2 shall be met with the specified line pairs per millimeter positioned at, and projected from the focal plane of the objective system.

3.11.4.1 On-axis. - The objective system shall resolve 200 line pairs per millimeter on axis.

3.11.4.2 Off-axis 0.35 inch. - The objective system shall resolve 200 line pairs per millimeter off-axis 0.35 inch.

3.11.5 Objective T number. - The requirements of 3.11.5.1 and 3.11.5.2 shall be met with the source of radiance at the focal plane of the objective system. T number is defined as the $\frac{F}{D}$ (ratio of focal length to equivalent diameter) of an idealized circular lens which:

a. Has no obscuration or transmission loss.
b. Has the equivalent focal length of the system in question.
c. Delivers the same irradiance at the focal plane as the system in question.

3.11.5.1 On-axis. - The T number of the objective system shall be not more than as-observed on-axis.

3.11.5.2 Off-axis 0.35 inch. - The T number of the objective system shall be not more than 3.20 as observed off-axis 0.35 inch.

3.11.6 Resolution - unity power system. - Resolution of the unity power system shall be 1.0 minute or less of arc when observed through any area of the exit pupil with a 3 power telescope having a 0.50 inch aperture.

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3.12 Optical characteristics, with image tube installed. - Unless otherwise specified, the requirements of 3.12.2 through 3.12.8.2 inclusive shall be met with:

a. The image tube installed in the periscope (see 3.12.1).
b. The periscope assembled to its mount assembly.
c. The clear filter positioned in the optical train.
d. The image tube, through the high voltage oscillator, and the reticle projector lamp energized (using 6 volts dc.)

3.12.1 Image intensifier tube, government furnished. - When the image intensifier tube is provided by the government each tube shall be subjected to verification by the contractor for the following requirements prior to insertion into the periscope.

3.12.1.1 Physical defects. - The image tube shall show no evidence of external or internal damage such as chips, digs, scratches, loose parts and other surface defects to fiber optics, glass and metal. The energized screen shall not contain pits, scratches or other surface defects in excess of tolerances specified in MIL-I-55536.

3.12.1.2 Voltage drop. - With an input voltage of 5.9 Vdc to the oscillator, and the oscillator output reading 2650 Vac (peak-to-peak), the image intensifier tube shall not cause a drop in voltage by more than 100 V (peak-to-peak) when the tube is placed in the circuit. There shall be no internal shorting, cathode fluttering or change in intensity nor bright or dark areas which could result in rejection of a complete periscope.

3.12.1.3 Luminance gain. - Luminance gain shall meet the requirements of MIL-I-55536.

3.12.1.4 Center resolution. - Center resolution shall meet the requirements of MIL-I-55536.

3.12.1.5 Image alinment. - Image alinement shall meet the requirements of MIL-I-55536.

3.12.2 Cleanliness.

3.12.2.1 Head assembly. - There shall be no visible evidence of condensates grease, fingerprints, and fractures when viewing the glass components by transmitting light from either end. In addition, the total number of dirt (or foreign) particles on the glass surface (combined) shall not exceed four (4) when viewing from either end. The maximum dimension of any particle shall be 0.025 inches.

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3.12.2.2 Unity power system - body assembly. - There shall be no visible evidence of condensates, grease, fingerprints or fractures when viewing the glass components, by transmitted light from either end. In addition, the total number of dirt (or foreign) particles on the glass surfaces (combined) shall not exceed six (6) when viewing from the eye end. The maximum dimension of any particle shall be 0.025 inches.

3.12.2.3 Nine power system.

a. With the image tube installed there shall be no more than six (6) particles of dirt (or foreign matter) appearing on the screen image when viewing from the eyepiece end of the periscope. The maximum apparent dimension of any particle shall not exceed the reticle line width. In addition, the minimum separation between particles shall represent 10 mils in object space. (excluding the image tube surface.)

b. With the image tube removed and a light source inserted in the tube cavity, there shall be no evidence of condensates, grease, fingerprints, fractures or adhesive separations when viewing the glass components from the objective and eyepiece end of the periscope. In addition, when viewing from the objective end, the total number of dirt (or foreign) particles on the glass surfaces (combined) shall not exceed four (4). The maximum apparent dimension of any particle shall be 0.025 inches.

3.12.3 Resolution - nine power system. - The nine power system shall be capable of resolving the following values (for the axis specified when observing a resolution target having the background luminance in foot lamberts (ft-1) and the contrast specified. Contrast (for the purpose of this requirement) is defined as the difference in density between the dark lines and the light background.

RESOLUTION TARGET		RESOLUTION	
Background Luminance	Contrast (density difference between black and clear)	On axis	OFF axis 2.0 degrees
1.5 x 10^{-2} ft-1	High 1.0 or greater	.65R_T	.65E_T
6 X 10^{-5} ft-1	Low 0.20 ± 0.5	.17R_T	

R_T is the minimum on axis resolution requirement for the image intensifier tube. The mandatory functional requirements of the Image Intensifier Tube Type 8586 are specified in MIL-I-55536.

3.12.4 Contrast transfer. - The following values (for the axis specified) shall be met with 2870° ± 100°K radiance, modulated at the special frequency (angular spacing) indicated, incident to the entrance aperture

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of the periscope. U_T is the minimum contrast transfer value obtained from the image intensifier tube assembly installed in the periscope being tested.

ANGULAR SPACING (in line pairs/milliradian) On axis	CONTRAST TRANSFER Vertical lines	Horizontal lines
.58	$.90U_T$	$.90U_T$
1.7	$.85U_T$	$.85U_T$
3.70	$.72U_T$	$.72U_T$

3.12.5 Luminance gain. - The luminance gain shall be not less than 525 foot lamberts.

3.12.6 Stray light luminance. - This requirement shall be met with the periscope set for infinity focus. Stray light luminance contribution over the image of a black disk, subtending a solid angle equal to the field of view; in a white surround, subtending a solid angle of not less than 120 degrees, shall not exceed 20 percent.

3.12.7 Field of view. - The field of view shall be not less than 90 milliradians.

3.12.8 Projected reticle pattern.

3.12.8.1 Definition and visibility, clear, medium and dark filters. - The image of the reticle pattern shall be well defined and clearly visible when viewed with the unaided eye under the following conditions:

a. The entrance window covered to prevent entrance of light using the clear and medium filter positions.
b. The periscope viewing an evenly illuminated field of not less than 10 foot lamberts luminance with the filter wheel in the dark position.

3.12.8.2 Reticle accuracy. - The angular value between the boresight cross and the maximum range graduation of the projected reticle pattern shall be within 2.0 percent of the value specified on the reticle drawing.

3.12.9 Accuracy. - Unless otherwise specified the requirements of 3.12.9.1.1 through 3.12.9.1.2, 3.12.9.2 through 3.12.9.5.3 and 3.12.9.7 through 3.12.9.8.4 shall be met with:

The conditions of 3.12a, 3.12b, 3.12c and 3.12d.
Datum Plane "A" defined as Datum Plane "A" shown on Drawing F10513500, horizontal.

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c. A plane defined by the two pads on Mount Assembly 10513500 (for the purpose of these requirements referred to as Datum Plane "D") perpendiculars
d. Surface "C" of the input coupling, shown on Drawing F10542204, positioned 67 degrees 30 minutes with respect to Datum Plane "A" as measured in a plane perpendicular to Datum Planes "A" and "D".
e. Elevation defined as the direction of line of sight deviation which occurs when the angular value defined in 3.12.9d is increased.

3.12.9.1 Image steadiness.

3.12.9.1.1 Flicker. - This requirement shall be met subsequent to allowing a 15 second time lapse to permit stabilization of the image tube. There shall be no evidence of flicker or movement of the image.

3.12.9.1.2 Sparking. - This requirement shall be met subsequent to allowing a 15 second time lapse to permit stabilization of the image tube. There shall be no evidence of sparking or arcing.

3.12.9.2 Collimation. - The requirements of 3.12.9.2.1 and 3.12.9.2.2 shall be met with the angular value specified in 3.12.9d changed to 66 degrees.

3.12.9.2.1 Elevation. - A line of sight which is originally perpendicular to Datum Plane "D" shall emerge parallel to Datum Plane "D" within 0.5 mils.

3.12.9.2.2 Deflection. - A line of sight which is originally perpendicular to Datum Plane "D" shall emerge perpendicular to the intersection of Datum Planes "A" and "D" within 1.0 mil.

3.12.9.3 Image tilt. - The requirements of 3.12.9.3.1 and 3.12.9.3.2 shall be met with the angular value specified in 3.12.9d changed to 66 degrees.

3.12.9.3.1 Unity power system. - The image of a target line which is originally perpendicular to Datum Plane "A" shall emerge perpendicular to Datum Plane "A" within 2.0 degrees of arc.

3.12.9.3.2 Nine power system. - The image of a target line which is originally perpendicular to Datum Plane "A" shall emerge perpendicular to Datum Plane "A" within 2.0 degrees of arc.

3.12.9.4 Parallelism of reticle and image. - The image of a target line which is originally parallel to Datum Plane "A" shall be parallel within 1.0 degree of arc to the 600 meter range line.

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3.12.9.5 Boresight movements. - The requirements of 3.12.9.5.1, 3.12.9.5.2 and 3.12.9.5.3 shall be met by the azimuth and elevation boresight movements.

3.12.9.5.1 Excursion range. - This requirement shall be met with the boresight cross originally in coincidence with an incident line of sight which is perpendicular to Datum Plane "D". The boresight movements shall provide a minimum of 6.0 mils movement in each direction simultaneously as read on the boresight scales.

3.12.9.5.2 Reticle shift. - Changing the direction rotation of the boresight knobs, through the range specified in 3.12.9.5.1 shall not cause the reticle to deviate more than 0.10 mil in a direction perpendicular to the intended movement.

3.12.9.5.3 Backlash. - The boresight movements in azimuth and elevation shall be capable of meeting this requirement over the full limit of travel in each direction. Backlash in each boresight movement shall not exceed 0.10 mil.

3.12.9.6 Input coupling.

3.12.9.6.1 Excursion range. - Starting with the input coupling positioned at the angular value specified in 3.12.9d rotation of the input coupling shall cause the line of sight to elevate a minimum of 22.0 degrees and depress a minimum of 18.0 degrees.

3.12.9.6.2 Input coupling travel. - Rotation of the input coupling shall be limited by contact with the travel stop pin.

3.12.9.6.3 Plumb travel. - Starting with the input coupling positioned at the angular value specified in 3.12.9d and the line of sight in coincidence with a target line which is perpendicular to Datum Plane "A", the line of sight shall track the target line within 0.25 mil when elevated and depressed through the range specified in 3.12.9.6.1.

3.12.9.6.4 Elevation accuracy. - This requirement shall be met with backlash removed and the angle measured In a plane which is perpendicular to Datum Planes "A" and "D". The angle formed by the line of sight and Surface "C" of the input coupling shall remain constant within 0.20 mil at any elevation setting through the range specified in 3.12.9.6.1.

3.12.9.6.5 Backlash. - Backlash between the line of sight and the input coupling shall not exceed 0.05 mil.

3.12.9.7 Dioptral adjustment.

3.12.9.7.1 Zero setting. - The diopter scale shall indicate zero within 0.25 diopter when the screen of the image tube is brought into best focus by eyepiece adjustment when viewing with a dioptometer.

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3.12.9.7.2 Excursion range - The diopter movement shall be adjustable through a minimum range of -3.5 to +3.5 diopters from the zero diopter scale position of the eyepiece as read on a dioptometer when viewing the screen of the image tube.

3.12.9.7.3 Backlash. - Backlash in the diopter movement shall not exceed 1/8 diopter.

3.12.9.8 Focus.

3.12.9.8.1 Range. - Rotation of the focusing knob shall bring targets from 40 feet to infinity into focus within the limits of the focusing knobs mechanical stops.

3.12.9.8.2 Image shift due to range focus knob. - Rotation of the range focus knob by 4° ±1°, clockwise and countercloehise, from the best focus position at 1500 meters shall not cause the target image to shift more than 3 mils (total spread) relative to an auxiliary telescope at the eyepiece of the periscope.

3.12.9.8.3 Effect on diopter setting. - The diopter setting shall not change more than 0.25 diopter, as read on the diopter scale, when the focusing knob is rotated through the range specified in 3.12.9.8.1.

3.12.9.8.4 Image centering. - With an incident line of sight perpendicular to Datum Plane "D" and the input coupling set at 66° with respect to Datum Plane "A", the target shall appear centered relative to the eyepiece diaphragm within 10 mils.

3.13 Operability.

3.13.1 Disengagement of knob locks. - The direct force required to disengage the diopter knob and each boresight knob shall be within the limits of 2.0 to 5.0 pounds (lbs.), and shall not exceed 9.0 lbs. at -65° and +150°F.

3.13.2 Torque. - The running torque required to rotate the controls listed below, through their full excursion of travel, shall be within the limits for the temperature indicated. The boresight slip scales and the filter lever shall be manually operable at all of the specified temperatures.

Control	60° to 90°F	-65° and +150°F
Boresight knobs	2.0 - 5.0 in.-lbs.	8.0 in.-lbs. max.
Diopter knob	3.0 - 7.0 in.-lbs.	14.0 in.-lbs. max.
Focusing knob	1.5 - 8.0 in. lbs.	15.0 in.-lbs. max.
Input coupling	3.0 - 7.0 in. lbs.	11.0 in.-lbs. max.

3.14 Sealing.

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a. Periscope. - The periscope assembled in accordance with 3.9a shall withstand an internal pressure of 5 ± 0.10 pounds per square inch gage (psig) for a minimum of 45 minutes with no evidence of leakage.

b. Head and body assembly with shipping covers installed. - The head and body assembly, with shipping covers, shall withstand an internal pressure of 5 ± 0.10 pounds per square inch gage (psig) for a minimum of 45 minutes with no evidence of leakage.

3.15 Flushing. - Upon satisfactory completion of sealing requirements of 3.14, the accepted head assembly and body assembly, with shipping covers installed, shall be flushed with dry nitrogen. The flushing shall continue until the gaseous mixture bled from the head and body assembly has a dew point of at least -25°F. The units shall then be sealed with dry nitrogen to an internal pressure of 3 psi ± .5 psi.

3.16 Cleanliness. - Upon satisfactory completion of flushing requirements of 3.15, the head assembly and body assembly shall meet the cleanliness requirements of 3.16.1 and 3.16.2.

3.16.1 Head assembly. - There shall be no visible evidence of condensates or fractures when viewing the glass components by reflected light at the front window of the head assembly. The total number of dirt (or. foreign) particles on the glass surfaces (combined) shall not exceed six (6). The maximum dimensions of any particle shall be 0.025 inches.

3.16.2 Body assembly, nine power system. - The requirements of 3.12.2.3a shall apply with the total number of dirt particles visible from the eyepiece end being increased from six (6) to eight (8).

3.17 Interchangeability.

3.17.1 Interchangeability - mechanical. - The periscopes (head and body assembly) 11731100 and 10542200 (see 2.1) shall be capable of accepting Mount Assemblies 10513360.

3.17.2 Interchangeability - pressure retention. - The periscopes (head and body assembly) 11731100 and 10542200 shall withstand an internal pressure of 5 ± 0.10 pounds per square inch for a minimum of two hours without loss of pressure in excess of .2 psig while mated to each Mount Assembly 10513360.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any

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other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) aproval. - The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) periscopes plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular productions

4.2.1 Government testing. - When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirewnts of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for meterials shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product. - Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. - All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual

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characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination and tests for pac~ging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 0.65%		
101. Line of sight, orientation, pre-vibration	3.9	4.6.3
102. Vibration	3.9	4.6.4, 4.6.6
103. Line of sight retention, post-vibration	3.9	4.6.5
104. Dielectric strength	3.10	4.6.7
105. Luminous transmittance, head assembly	3.11.3a	4.6.8.7
106. Resolution, head assembly	3.11.3b	4.6.8.8
107. Resolution, unity power system	3.11.6	4.6.8.13
108. Cleanliness, head assembly	3.12.2.1	4.6.10.1
109. Cleanliness, unity power system	3.12.2.2	4.6.10.2
110. Cleanliness, nine power system tube Installed	3.12.2.3a	4.6.10.3
111. Cleanliness, nine power system, tube removed	3.12.2.3b	4.6.10.4
112. Resolution, nine power system on axis	3.12.3	4.6.10.5
113. Field of view	3.12.7	4.6.10.10
114. Projected reticle pattern, definition and visibility	3.12.8.1	4.6.10.11
115 . Reticle accuracy	3.12.8.2	4.6.10.12
116. Flicker, image steadiness	3.12.9.1.1	4.6.11.1
117. Sparking, image steadiness	3.12.9.1.2	4.6.11.2
118. Collimation, elevation	3.12.9.2.1	4.6.11.3
119. Collimation, deflection	3.12.9.2.2	4.6.11.3
120. Image tilt, unity power system	3.12.9.3.1	4.6.11.4
121. Image tilt, nine power system	3.12.9.3.2	4.6.11.5
122. Parallelism of reticle and image	3.12.9.4	4.6.11.6
123. Excursion range, boresight movements	3.12.9.5.1	4.6.11.7
124. Reticle shift, azimuth and elevation boresight movement	3.12.9.5.2	4.6.11.8
125. Backlash, azimuth and elevation boresight movement	3.12.9.5.3	4.6.11.9
126. Elevation accuracy, input coupling	3.12.9.6.4	4.6.11.10

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TABLE I - CLASSIFICATION OF DEFECTS CONT'D

		REQUIREMENTS	TEST PROCEDURE
127.	Excursion range, input coupling	3.12.9.6.1	4.6.11.11
128.	Plumb travel	3.12.9.6.3	4.6.11.12
129.	Backlash, input coupling	3.12.9.6.5	4.6.11.13
130.	Input coupling travel	3.12.9.6.2	4.6.11.14
131.	Zero setting of diopter scale	3.12.9.7.1	4.6.11.15
132.	Excursion range, diopter movement	3.12.9.7.2	4.6.11.16
133.	Backlash, diopter movement	3.12.9.7.3	4.6.11.17
134.	Focus range	3.12.9.8.1	4.6.11.18
135.	Image shift due to range focus knob	3.12.9.8.2	4.6.11.19
136.	Focus effect on diopter setting	3.12.9.8.3	4.6.11.20
137.	Image centering	3.12.9.8.4	4.6.11.21
138.	Operability, disengagement of knob locks	3.13.1	4.6.12
139.	Operability, boresight knobs torque	3.13.2	4.6.13
140.	Operability, diopter knob torque	3.13.2	4.6.13
141.	Operability, focusing knob torque	3.13.2	4.6.13
142.	Operability, input coupling torque	3.13.2	4.6.13
143.	Sealing, assembled persicope	3.14a	4.6.14
144.	Sealing, head assembly with shipping cover installed	3.14b	4.6.15
145.	Sealing, body assembly with shipping cover installed	3.14b	4.6.15
146.	Flushing, head and body assembly	3.15	4.6.16
147.	Cleanliness, head assembly	3.16.1	4.6.17
148.	Cleanliness, body assembly nine power system	3.16.2	4.6.18
149.	Interchangeability - Mechanical	3.17.1	4.6.19.1
150.	Workmanship	3.6	Visual-Tactile

MINOR: NONE

NOTE: The tests in Table I shall be conducted at a temperature between +60° and +90°F.

4.3.3 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

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4.4 Special sampling

4.4.1 General. - One periscope shall be selected at randon from each 100 produced or from each month's production whichever occurs first. The sample shall meet the requirements and tests in Table II.

TABLE II

NO.	CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURE
301.	Fabrication	3.1	Applicable Drawings - visual
302.	General specification	3.5	MIL-F-13926 - visual

4.4.2 Environmental. - Three periscopes shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table 111 and shall then meet the requirements and tests in Table I with exception of Characteristics 101, 102 and 103.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURE
303.	Line of sight orientation	3.9	4.6.3
304.	Extreme storage temperatures (-80°F to +160°F)	3.7	4.6.1
305.	Operability, disengagement of knob locks (-65°F and +150°F)	3.8, 3.13.1	4.6.2
306.	Operability, torque (-65°F and +150°F)	3.8, 3.13.2	4.6.2
307.	Vibration	3.9	4.6.4, 4.6.6
308.	Line of sight retention, post vibration	3.9	4.6.5
309.	Contrast transfer off-axis objective system	3.11.1.1b	4.6.8.2
310.	Contrast transfer off-axis, eyepiece and erecting system	3.11.1.2b	4.6.8.3
311.	Luminous transmittance, objective system	3.11.2.1a	4.6.8.4
312.	Luminous transmittance at 0.85 micron, objective system	3.11.2.1b	4.6.8.5
313.	Luminous transmittance, eyepiece and erecting system	3.11.2.2	4.6.8.6
314.	Resolution on-axis, objective system	3.11.4.1	4.6.8.9

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TABLE III CONT'D

NO.	CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURE
315.	Resolution off-axis, objective system	3.11.4.2	4.6.8.10
316.	Objective T number on-axis	3.11.5.1	4.6.8.11
317.	Objective T number off-axis	3.11.5.2	4.6.8.12
318.	Resolution, nine power system off-axis	3.12.3	4.6.10.6
319.	Stray light luminance	3.12.6	4.6.10.9
320.	Interchangeability - pressure retention	3.17.2	4.6.19.2

NOTE: The tests for Characteristics 303 and 307 through 32o inclusive, shall be conducted at a temperature between +60° and +90°F.

4.4.3 One hundred Percent (100%) inspection. - All periscopes produced, including those subjected to the requirements and tests in Tables I and III, shall meet the requirements and tests in Table IV.

TABLE IV

NO.	CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURE
320.	Contrast transfer, on-axis objective system	3.11.1.1a	4.6.8.1
321.	Contrast transfer, on-axis eyepiece and erecting system	3.11.1.2a	4.6.8.3
322.	Contrast transfer, on-axis (image tube installed)	3.12.4	4.6.10.7
323.	Luminance gain	3.12.5	4.6.10.8

NOTE: The tests for Characteristics 320 through 323 inclusive, shall be conducted at a temperature between +60° and +90°.

4.4.4 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment. Except as otherwise provided for by the contracct, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

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4.5.1 Government furnished inspection equipment. - Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. - Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design. - The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Orientation, vibration (pre-vibration and post-vibration line of sight). - The special testing equipment shall consist of the following main assemblies:

a. Mounting plate assembly
b. Line of sight test assembly
c. Vibration test stand assembly
d. Vibrating machine
e. Power source

4.5.2.3.1.1 Mounting plate assembly. - The mounting plate assembly shall incorporate the necessary locating and locking features of the Mount Assembly 10513500. The mounting plate assembly shall permit mounting and orienting the periscope as specified in 3.9. A device shall be incorporated on the mounting plate assembly that positions and solidly arrests surface "C" of the periscope input coupling, as shown on Drawing F10542204, 67 degrees 30 minutes ± 10 seconds with respect to Datum Plane "A" of the head assembly. Features shall be designed into the mounting plate assembly that will assure positive and accurate positioning and repositioning (repeated removal and replacement) of the mounting plate

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assembly on the line of sight test stand assembly within 5 seconds in all planes.

4.5.2.3.1.2 Line of sight test stand assembly. - The assembly shall function as a holder for the mounting plate assembly with the persicope held in its normal operating position. Features shall be designed into the line of sight test stand assembly that will assure positive and accurate positioning and repositioning (repeated removal and replacement) of the mounting plate assembly on the test stand assembly within 5 seconds in all planes. A target projector collimator having a clear aperture of at least 7 inches, with an appropriate crossline reticle patter shall also be part of the test stand assembly. The target projector collimator shall be mounted so that the optical axis of the collimator is in alinement with the line of sight emitting from the objective end of the periscope. The target projector collimator shall be set at infinity focus, unless otherwise specified. A set-up device of the front surface reflecting mirror type shall be utilized for the alinement (auto-collimation) of the target projector collimator. The optical axis of the target projector collimator shall be alined (auto-collimated) parallel with respect to Datum Plane "A" of the periscope and perpendicular with respect to Datum Plane "B" of the periscope within 5 seconds. The above set-up device and all additional set-ups for testing equipment, with complete provisions necessary for certification and proof of accuracy of the test equipment shall be provided and will become part of the test equipment.

4.5.2.3.1.3 Vibration test stand assembly. - The vibration test stand assembly shall function as a holder for the mounting plate assemly with the periscope in it's normal operating position. Features shall be designed into the vibration test stand assembly that will assure positive and secure mounting of the mounting plate assembly on the vibration test stand. The vibration test stand shall be firmly and secured mounted on a vibration machine.

4.5.2.3.1.4 Vibrating machine. - The vibrating machine shall be capable of providing the amplitude and frequency specified in 3.9d. The vibrating machine shall also be capable of vibrating the periscope along 3 mutually perpendicular axes while the periscope is maintained in the position specified in 4.5.2.3.1.3.

4.5.2.3.1.5 Power source. - The regulated power supply specified in 4.5.3 shall be utilized to energize the persicope oscillator and reticle projector. The primary power input of 5.9 $\pm$ .1 VDC shall be applied in positive polarity to the applicable connectors of the Cable Assembly 8589783, terminals on the periscope body assembly and to connect the oscillator contact and reticle projector contact to the regulated power supply.

4.5.2.3.2 Accuracy. - The special testing equipment utilized for the following tests shall conform to 4.5.2.3.2.1, 4.5.2.3.2.2, 4.5.2.3.2.3 and 4.5.2,3.2.4.

a. Collimation, elevation and deflection
b. Image tilt, unity power and nine power systems
c. Parallelism of reticle and image
d. Boresight movements, excursion range
e. Reticle shift and backlash

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f. Excursion range, input coupling
g. Coupling travel, input coupling
h. Plumb travel
i. Elevation accuracy and backlash

4.5.2.3.2.1 Orientation and mounting of the Periscope. - The special testing equipment shall consist of a mounting plate assembly that incorporate the necessary locating and locking features of the Mount Assembly 10513500. The mounting plate assembly shall permit mounting and orienting the periscope as specified in 3.12.9. A test stand assembly, capable of positioning and securing the mounting plate assembly with periscope shall be a part of the test equipment. Adjustments shall be provided on the test stand assembly to permit orientation of Datum Plane "A" in a horizontal plane within 10 seconds. The test equipment shall also consist of a device that is capable of positioning and solidly arresting Surface "C" of the periscope coupling, as shown on drawing F10542204, 67 degrees 30 minutes ± 10 seconds with respect to Datum Plane "A" of the mount. This device, subsequent to the above positioning, shall also be capable of articulation of Surface "C" to any angular position between its travel stops. A front surface reflecting mirror, capable of adjustment in all planes, mounted directly on the periscope coupling positioning device, shall be positioned so that the reflecting surface of the reflecting mirror is in alinement with the line of sight emitting from the objective end of the periscope. A calibrated penta prism shall be utilized in the line of sight between a viewing telescope and head assembly. When the head assembly (less body assembly) is inspected for the requirements of 3.12.9.2 and 3.12.9.6, the testing equipment for the head assembly tests shall be accurate within 5 seconds.

4.5.2.3.2.2 Target collimator. - A target projector collimator, with a clear aperture of at least 7 inches, with an appropriate test reticle pattern, shall be a part of the test equipment. The target projector collimator shall be positioned and mounted so that the optical axis of the collimator is in alinemant with the line of sight emitting from the objective end of the periscope. The target projector collinuator shall be set at infinity focus, unless otherwise specified. The target projector collimator test reticle shall be within the accuracies indicated for the following requirements:

Reticle shift Backlash (boresight movements)	Five seconds of arc
Parallelism of reticle and image	Six minutes of arc
Image centering	1 mil
Unity power system Nine power system	Twelve minutes of arc

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4.5.2.3.2.3 Viewing telescope. - A viewing telescope, capable of measuring any line of sight deviations at the periscope eyepiece, in the "X" and "Y" coordinates in seconds of arc shall be a part of the test equipment. The viewing telescope shall be mounted so that the optical axis of the telescope is in alinement with the line of sight emitting from the eyepiece of the periscope.

4.5.2.3.2.4 Set-up device. - A set-up device of the front surface reflecting mirror type shall be utilized for the alinement (auto-collimation) of the periscope coupling articulation device front surface reflecting mirror, the target projector collimator, and the viewing telescope. The reflecting surface of the periscope coupling articulation device-front surface reflecting mirror shall be alined (auto-collimated) perpendicular with respect to Datum Plane "A" and parallel with respect to Datum Plane "D" (see 3.12.9c) within 5.0 seconds. The optical axis of the target projector collimator shall be alined (auto-collimated) parallel with respect to Datum Plane "A" and perpendicular to Datum Plane "D" within 5.0 seconds. The optical axis of the viewing telescope shall be alined (auto-collimated) parallel with respect to Datum Plane "A" ati perpendicular to Datum Plane "D" within 5.0 seconds. The optical axis of the viewing telescope shall be alined (auto-collimated) parallel with respect to Datum Plane "A" and perpendicular to Datum Plane "D" within 5.0 seconds. The set-up device shall also be capable of alining the mating coupling of the periscope coupling articulation device at 67 degrees 30 minutes within 10 seconds with respect to Datum Plane "A". All additional set-up and calibration equipment, with complete provisions necessary for certification and proof of accuracy of the test equipment shall be provided and shall become part of the testing equipment.

4.5.2.3.3 Image intensifier tube. - The testing equipment shall conform to the applicable test equipment requirements of MIL-I-55536 for testing luminance gain, center resolution, and image alinement of the image intensifier tube. The testing equipment shall provide the necessary features for monitoring the voltages utilized in the tests of the imge tube.

4.5.2.3.4 Sealing - periscope. - Special testing equipment to test sealing of the periscope shall conform to the internal pressure test requirements of Sheet 1 of Drawing F8565556, Pressure Tester-Two Station, except that the regulator shall conform to the requirements for use of dry air.

4.5.2.3.5 Sealing - head and body assemblies. - Special testing equipment to test sealing of the head and body assemblies shall conform to the internal pressure test requirements of Sheet 1 of Drawing F8565556, Pressure Tester-Two Station.

4.5.2.3.6 Force requirements (disengagement). - Testing equipment for measuring direct force (pull) with a minimum of zero to twelve pounds,

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the accuracy of which shall be plus or minus two ounces throughout the entire zone of excursion, shall be utilized. The testing equipment shall be equipped with suitable adapters to accommodate testing of the diopter knob and boresight knobs.

4.5.2.3.7 Torque requirements. - A torque wrench type tester with a minimum range from zero to twenty inch-pounds, the accuracy of which shall be plus or minus two inch ounces throughout the entire zone of excursion, shall be utilized for the torque tests. The testing equipment shall be equipped with suitable adapters to accommodate testing of the boresight knobs, diopter knob, focusing knob and input coupling.

4.5.3 Test equipment. - In conjunction with 4.5.2.3 the following standard test equipment shall be utilized in the performance of the applicable tests as specified in 4.6:

NOMENCLATURE	DESCRIPTION
1. Regulated D.C. Voltage Supply	5-7 VDC output, regulated .015%, .1% accuracy of full scale reading. Ripple less than 250 micro volts (RMS) or 1 milli-volt peak-to-peak.
2. Automatic DC Digital Voltmeter	Accuracy: ±2% of reading.

4.5.4 Set-up and calibration. - Rigidity, stability, ease of maintenance, and adaptability of set-up and calibration equipment with complete provisions are also prime requirements. All set-up and calibration equipment necessary for certification of the testing equipment accuracy shall be furnished and will become a part of the testing equipment.

4.6 Test methods and procedures.

4.6.1 Extreme storage temperature. - The periscope subjected to this test shall not be assembled to its mount, but shall be tested as a head assembly and body assembly. These assemblies shall have their respective shipping covers installed. The head assembly and body assembly shall be pressurized with dry nitrogen gas to an internal pressure of 3 ± 0.5 psig. The nitrogen gas shall have a dew point at least as low as -25°F. The head assembly and body assembly (periscope) shall be exposed to the temperatures specified in 3.7 utilizing testing equipment in accordance with the "Test Facilities" requirements of MIL-F-13926. The head assembly and body assembly shall be placed in the test chamber and the temperature of the chamber reduced gradually (see Note 1) to -80°F. The head assembly shall remain at this temperature for a minimum of 2 hours. The body assembly shall remain at this temperature for a minimum of 5 hours. At the completion of the applicable 2 hour and 5 hour periods, the surrounding temperature of the head and body assemblies shall be gradually raised to

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+160°F. The head and body assemblies shall remain at this temperature for a minimum of 2 hours and 5 hours respectively. At the completion of the respective 2 hour and 5 hour periods the temperature of the test chamber shall be gradually reduced to room ambient temperature (+60° to +90°F), and the head assembly and body assembly removed from the test chamber. The shipping covers shall be removed from the head assembly and body assembly. The head assembly, body assembly and shipping covers shall then be subjected to the visual, tactile and audible examination of 4.6.6. The head assembly, body assembly and shipping covers shall show no evidence of physical failure as specified in 3.7. Caution shall be exercised during this test to avoid subjecting the head assembly and body assembly to thermal shock.

NOTE 1: The rate of temperature change in the test chamber shall not exceed 20 degrees per hour throughout the temperature cycling tests of 4.6.1 and 4.6.2.

4.6.2 Extreme operating temperature. - The head assembly and body assembly subjected to this test shall have its shipping cover installed. The head assembly and body assembly shall be pressurized with dry nitrogen gas to an internal pressure of 3 ± 0.5 psig. The nitrogen gas shall have a dew point at least as low as -25°F. The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.8. The head assembly and body assembly shall be placed in the test chamber and the temperature of the chamber reduced gradually (see Note 1) to -65°F. The head assembly shall remain at this temperature for a minimum of 2 hours. The body assembly shall remain at this temperature for a minimum of 5 hours. At the completion of the applicable 2 hour and 5 hour period, and while at -65°F, the head assembly and body assembly shall be subjected to the test in 4.6.12 and 4.6.13. Upon completion of this test the surrounding temperature of the head and body assemblies shall be gradually raised to +150°F. The head and body assemblies shall remain at this temperature for a minimum of 2 hours and 5 hours respectively. At the completion of the applicable 2 hour and 5 hour period, and while at +l50°F, the head assembly and body assembly shall be subjected to the test in 4.6.12 and 4.6.13. Upon completion of this test the temperature of the test chamber shall be gradually reduced to room ambient temperature (+60° to +90°F) and the head assembly and body assembly removed from the test chamber. The head and body assemblies shall conform to the requirements of 3.8. Caution shall be exercised during this test to avoid subjecting the head assembly, body assembly and shipping covers to thermal shock.

4.6.3 Line of sight orientation pre-vibratione. - The periscope shall be positioned as specified in 3.9 utilizing special testing equipment conforming to 4.5.2.3.1. Energize the periscope. Set the focusing knob of the periscope for the appropriate range setting for this test. Adjust the diopter knob of the periscope for best focus of the phospor grain on the screen of the image tube, while viewing through the periscope eye-piece. Adjust the periscope reticle for best intensity. Obtain coincidence

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of the horizontal and vertical reticle lines of the periscope and the target image crosslines by means of the periscope boresight knobs. While maintaining the coincidence of reticle lines and target image lines, slip the scales of the boresight knobs to their nominal "zero". Subsequent to setting the boresight knob scales to zero, remove the mounting plate assembly (4.5.2.3.1.1) with periscope from the line of sight test stand (4.5.2.3.1.2). Position the mounting plate assembly with periscope on the vibration test stand (4.5.2.3.1.3) and subject the periscope to the test in 4.6.4. Caution must be exercised during this transfer of the mounting plate assembly. All unnecessary shocks which would tend to render the testing equipment inaccurate for subsequent tests shall be avoided.

4.6.4 Vibration of Periscope. - Perform this test with the periscope and equipment positioned, as specified in 3.9, on special testing equipment that conforms to 4.5.2.3.1.4. Energize the periscope, oscillator and reticle projector. The periscope shall be vibrated in accordance with Procedure IX, Part 1, of MIL-STD-810, except that the frequency, frequency cycling and amplitude of vibration shall be as specified in 3.9. During the vibration cycle the periscope shall be subjected to a visual examination. The periscope shall conform to the requirements of 3.9e and 3.9f. Immediately following the vibration cycle, and prior to the examination for evidence of physical failure the periscope shall be subjected to the test in 4.6.5. Caution shall again be exercised during transfer of the mounting plate assembly with the periscope. Upon completion of the test in 4.6.5 the periscope shall be subjected to the tests specified in 4.6.6.

4.6.5 Line of sight retention - post vibration. - Conduct this test subsequent to vibration, and prior to the examination for physical failure resulting from vibration. Position the mounting plate assembly (4.5.2.3.1.1) with the periscope on the line of sight test stand as in 4.6.3. Inspect the periscope boresight knobs for any evidence of rotation from their "zero" setting established in 4.6.3. Observe the periscope horizontal and vertical reticle lines with respect to the target image crosslines. Any lack of coincidence between the periscope reticle lines and the target image lines shall not exceed the tolerance specified in 3.9g, as measured on the periscope boresight knobs.

4.6.6 Examination. - Subsequent to vibration the periscope shall be thoroughly examined for any evidence of physical failure. A visual and tactile examination shall be made for missing or damaged parts. A tactile examination shall also be conducted to determine looseness or improper functioning of moving parts, where applicable. An audible examination shall be made of the periscope by shaking it and listening for loose internal parts or broken components. Any one of the possible defects defined above shall be cause to consider any assembly defective. The periscope shall be subjected to the tests in 4.6.7 through 4.6.13 inclusive to determine overall compliance to 3.9.

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4.6.7 Dielectric strength. - Perform this test in accordance with Method 301 of MIL-STD-202, utilizing standard measuring equipment. Dielectric strength shall be in conformance with the requirements of 3.10. This is an in-process test that is performed with the components specified in 3.10 removed from the circuitry. The inspection records for each periscope shall indicate the results of this test.

4.6.8 Optical characteristics, with image tube removed. - Position the periscope on a holding device that incorporates the necessary holding features of the mounting plate assembly specified in 4.5.2.3.1.1. Unless otherwise specified, the image tube shall be removed from the periscope and the clear filter shall be positioned in the optical train. The test values for all characteristics shall be recorded and be made part of the inspection record.

NOTE: Unless otherwise specified the inspection of the periscope shall be conducted in an atmosphere that is controlled for temperature, humidity, and contaminates that would adversely affect the head assembly or body assembly when separated and not assembled as a periscope.

4.6.8.1 Contrast transfer on-axis, objective system.- Perform this test using a 2870° ± 100 K Kelvin (K) light source, diffuse screen, frequcncy pattern (see Figure 1), collimator,an S-20 extended red photo. detector with slit (see Figure 2), and an "X-Y" recorder set up and oriented as in Figure 3. Position the photo-detector in the tube cavity of the periscope utilizing an appropriate adapter. Focus the image of the frequency pattern at the slit. Adjust the "X-Y" recorder for zero level by blocking the entrance of all light to the photo-detector. Measurements shall be made at 5 frequencies in the image plane. Position the photo-detector on-axis and advance the frequency pattern, in a horizontal direction, through the five frequencies at a constant rate such that no signal damping due to response time of the photo-detector occurs, and obtain recordings similar to that shown in Figure 4. Calculate the on-axis contrast transfer for vertical lines using the formula $\frac{W - B}{W + B}$ from measured data. Record the contrast transfer values for the vertical lines. Rotate the frequency pattern and slit 90%. Measure, calculate, and record the on-axis contrast transfer values for the horizontal lines. Contrast transfer on-axis shall conform to the requirements specified in 3.11.1.1a.

NOTE: Special care must be taken before test data is made to avoid possible introduction of contrast transfer losses resulting from the instrument.

4.6.8.2 Contrast transfer off-axis, objective system. - Perform this test in sequence to the test in 4.6.8.1 utilizing the testing equipment and test method specified therein. Off-axis measurements shall

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be accomplished by translating the microscope objective and slit (Figure 2) perpendicular to the optical axis without refocusing Measure, calculate, and record the off-axis contrast transfer values for the vertical lines with the microscope objective and slit translated off-axis to the left from on-axis. Translate the microscope objective and slit downward from on-axis and repeat the measurements, calculations, and recording for both vertical lines and horizontal lines. Contrast transfer off-axis values shall conform to the requirements specified in 3.11.1.1b (see 4.6.8,1 NOTE).

4.6.8.3 Contrast transfer on and off-axis, eyepiece and erecting system. - Perform this test using a 2870° ± 100K light source, diffuse screen, frequency pattern (see Figure 1), collimator, an eye corrected photo-detector with slit (see Figure 5), and an "X-Y" recorder set up and oriented as in Figure 6. An appropriate adapter shall be utilized to position the photo-detector in the tube cavity of the periscope. Set the eyepiece of the periscope to zero diopters. Perform this test using the test methods and procedures specified in 4.6.8.1 and 4.6.8.2. Measure, calculate, and record the on-axis contrast transfer values for both vertical lines and horizontal lines. Measure, calculate, and record the off-axis contrast transfer values for the vertical lines with the microscope objective and slit translate off-axis to the right from on-axis. Translate the microscope objective and slit upward and repeat the measurements, calculations, and recording for both vertical lines and horizontal lines. Contrast transfer on-axis and off-axis shall conform to the applicable requirements specified in 3.11.1.2a and 3.11.1.2b (see 4.6.8.1 NOTE).

4.6.8.4 Luminous transmittance of objective system. - A diffuse surface with a luminance of not less than 100 foot lamberts shall be positioned at the image plane of the objective system (image tube cavity) of the periscope. The light source shall be from a 2870° ± 100°K tungsten lamp (see Figure 12). The entire entrance pupil of the head assembly shall be scanned with an eye corrected photometer. The luminous transmittance of the image, measured at the entrance pupil at any point, shall be not less than the value specified in 3.11.2.1a. Percentage of transmission shall be defined as:

$$\frac{\text{Apparent luminance as seen thru periscope}}{\text{Luminance of target}} \times 100\%$$

4.6.8.5 Luminous transmittance at 0.85 micron of objective system. - Perform this test in sequence to the test in 4.6.8.4. This test may be performed with either an image forming radiometer or an non-image forming radiometer used at a controlled distance. The radiometer shall be filtered for 0.85 micron. The filter utilized with the radioxneter shall have a peak transmission of not less than 30% at 0.85 micron, and shall have a band width of +0.10 - 0.01 micron at 1/3 of peak. When an image forming radiometer (detector assembly) is used, the test may be performed in either direction, that is, with the light source outside the periscope and the

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detector assembly in the tube cacity, or with the light source in the tube cavity and the detector assembly outside the periscope. In either case extreme care must be taken to assure that the detector assembly reads exactly the same angle when measuring the periscope objective system as when being calibrated at 100%. When an non-image forming detector is used in the tube cavity it must be capable of measuring all the energy which enters the periscope objective system and all the energy which reaches the tube cavity. The measured transmittance at 0.85 micron shall not be less than the value specified in 3.11.2.1b.

NOTE: A narrow band radiation source may be substituted for the 2870°K illuminance source in this test. The spectral response of the light source shall be equivalent to the spectral response of the filter.

4.6.8.6 Luminous transmittance - eyepiece and erecting system. - A diffuse surface with a luminance of not less than 100 foot lamberts shall be positioned in the object plane (image tube cavity) of the eye telescope. The luminance of the target as seen at the eyepiece of the periscope shall be not less than the value specified in 3.11.2.2 when measured with an eye corrected photometer.

4.6.8.7 Luminous transmittance - head assembly. - Conduct this test with the body assembly removed from the mounting plate assembly. A diffuse luminous source from 2870° ± 100°K tungsten shall be positioned at the exit aperture. The transmittance measurement shall be made with an eye corrected photometer while scanning the entrance aperture. The transmittance shall be not less than the value specified in 3.11.3a as measured on the photometer. Compute the percentage of transmittance using the following:

$$\frac{\text{Luminance of source as seen through head}}{\text{Luminance of source}} \times 100\%$$

4.6.8.8 Resolution - head assembly. - Perform this test utilizing a resolution target and a telescope of at least 90 power with an entrance pupil of at least 7.0 inches. The resolution target shall be placed on a wall target or in a collimator, positioned incident to the entrance window, and shall represent the angular subtense for the seconds of arc specified. Construction of the resolution target shall conform to the USAF 1951 Resolution Pattern in accordance with MIL-STD-150. The target shall be illuminated by diffuse light from a 2870° ± 100°K tungsten lamp. The resolution target shall have the detectable line structure and proper line count when observed, at one focus setting, through the telescope positioned at the exit aperture of the head assembly. Resolution shall meet the requirements of 3.11.3b.

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4.6.8.9 Resolution on-axis - periscope objective assembly. - Position a resolution pattern on a reticle placed in the focal plane of the objective system of the periscope (image tube cavity). Construction of the resolution pattern reticle shall be as shown in Figure 7. Project the resolution pattern by diffuse light from a 2870° ± 100°K tungsten lamp. The resolution pattern shall have the detectable line structure and proper line count when observed through a telescope positioned at, and normal to, the entrance aperture of the head assembly (see Figure 8). The telescope shall conform to the telescope utilized in 4.6.8.8. Resolution on-axis shall meet the requirements of 3.11.4.1.

4.6.8.10 Resolution off-axis - periscope objective assembly. - Perform this test as specified in 4.6.8.9, except that the viewing telescope shall be rotated off-axis the required distance. Resolution off-axis shall meet the requirements of 3.11.4.2.

4.6.8.11 Objective T-number - on-axis. - The effective T-number will be determined by flooding the image plane of the periscope objective with a known luminosity and measuring the illumination of the projected image on a target placed in front of the periscope. A luminous source providing a diffuse luminosity of approximately 100 foot lamberts over a circular area 1.0 inch in diameter shall be positioned in the image tube cavity by means of a suitable adapter. A reticle shall be mounted between the luminous source and the objective lens and positioned to provide a sharply focused image at a distance of 10.0 feet from the entrance window of the head assembly. The reticle pattern shall be as indicated in Figure 11. Luminance measurements shall be made from a magnesium oxide target placed within the center cross mark of the reticle image at 10.0 feet. The objective T-number shall not exceed the value specified in 3.11.5.1.

4.6.8.12 Objective T-number - off-axis. - Perform this test as in 4.6.8.11. The objective T-number shall be measured at the 4 peripheral cross marks of the reticle image. The objective T-numbers shall not exceed the value specified in 3.11.5.2.

4.6.8.13 Resolution - unity power system. - The test for resoltuion shall be performed utilizing an auxiliary telescope conforming to the requirements of 3.11.6, and a resolving power chart on a wall target or positioned in a collimator. The resolving power target shall represent the angular subtense for the minutes of arc specified. Construction of the target shall conform to the USAF 1951 Resolution Pattern in accordance with MIL-STD-150. The target shall be illuminated by diffuse light from a 2870° ± 100°K tungsten lamp. The target shall have the detectable line structure and proper line count when observed through the telescope positioned at the exit aperture of the unity power system. Resolution shall meet the requirements of 3.11.6.

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4.6.9 Image intensifier tube, Government furnished. - Each Government furnished image tube shall be subjected to the following verification inspections prior to insertion into the periscope:

4.6.9.1 Physical defects. - Prior to energizing the tube it shall be subjected to a visual and tactile examination. The tube shall show no evidence of physical defects as specified in 3.12.1.1. The image tube shall than be positioned on the testing equipment specified in 4.5.2.3.3 and energized. Visually examine the energized screen. The energized screen shall conform to the requirements of 3.12.1.1.

4.6.9.2 Voltage drop. - Position the image tube on the fixture specified in 4.5.2.3.3. Provisions shall be made on the fixture for the engagement and disengagement of the energized oscillator to the image intensifier tube. Energize the disengaged oscillator to the values specified. Engage the energized oscillator and image tube. The output voltage drop of the oscillator shall not exceed the value specified in 3.12.1.2 as indicated on the monitor of the oscillator output voltage. The tube shall not evidence any audible or visible defect identified in 3.12.1.2.

4.6.9.3 Luminance gain, center resolution and image alinement. - The examination and tests for these characteristics shall be in accordance with MIL-I-55536. The image tube shall conform to the requirements of 3.12.1.3 through 3.12.1.5 inclusive.

4.6.10 Optical characteristics with image tube installed. - The conditions, and positioning of the periscope, for these tests shall be as specified in 4.6.8, except that the image tube shall be installed in the periscope. The test equipment specified in 4.5.2.3.1.5 shall be utilized to energize the oscillator and reticle projector of the periscope.

4.6.10.1 Cleanliness, head assembly. - Perform this test prior to assembling the body assembly to the mounting plate assembly. The head assembly shall be visually examined through the entrance window and exit aperture with the unaided eye. Inspection for moisture shall be made by the technique of shadowing. Inspection for dirt or foreign particles shall be made by viewing a uniformly illuminated field having a brightness of approximately 300 apparent foot candles. Cleanliness of the head assembly shall conform to the requirements of 3.12.2.1.

4.6.10.2 Cleanliness of unity power system, body assembly. - Perform this test on the body assembly prior to its assembly to the mounting plate assembly. Position a diffuse light source at the entrance window, then visually inspect the unity power optical system through the exit window. Then position the light source at the exist window and visually inspect the optical system through the entrance window. The unity power system of the body assembly shall conform to the requirements of 3.12.2.2.

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4.6.10.3 Cleanliness of nine power system, tube installed. - Perform this test with the head assembly and body assembly assembled as a periscope. Present diffuse illumination from a 2870° ± 100°K tungsten lamp to the entrance window of the head assembly. Position the periscope switch in the ON position and project the periscope reticle into the optical system. Visually inspect the nine power system through the periscope eyepiece. The nine power system shall conform to the requirements of 3.12.2.3a.

4.6.10.4 Cleanliness of nine power system, tube removed. - Perform this test with the head assembly and body assembly assembled as a periscope, and the image tube removed from the body assembly. Position a diffuse white light source in the image tube cavity. Visually inspect the objective system through the head window, then the eyepiece system through the periscope eyepiece. The nine power system shall conform to the requirements of 3.12.2.3b.

4.6.10.5 Resolution - nine power system, on-axis. - Perform this test using a projected or mechanical target. The effective range shall be not less than 40 feet. The radiance presented to the periscope shall have a color temperature of 2870° ± 100°K. the background luminance and contrast (high and low) of the target shall be as specified in 3.12.3. Construction of the target shall conform to the USAF 1951 Resolution Pattern in accordance with MIL-STD-150. Adjust the focus of the periscope. The resolution target shall have the detectable line structure and proper line count when observed through the eyepiece with the unaided eye. Resolution on-axis for high and low contrast shall meet the requirements specified in 3.12.3.

CAUTION: The image tube must be erased (completely discharged), and the observer dark adapted for not less than 15 minutes before performing the low contrast resolution tests.

4.6.10.6 Resolution - nine power system, off-axis. - Rotate the periscope off-axis 2.0 degrees and repeat the test specified in 4.6.10.5 without changing the orientation of the focus knob from the position established in that test. Resolution off-axis for high contrast shall meet the requirements specified in 3.12.3.

4.6.10.7 Contrast transfer - on-axis. - Perform this test using a 2870° ± 100°K tungsten light source, diffuse screen, frequency pattern (see Figure 1), collimator, eye corrected photo-detector with slit, and an "X-Y" recorder set up and oriented as in Figure 9. Illuminate the frequency pattern by means of a 2870° ± 100°K tungsten lamp and focus the image of the frequency pattern at the slit. Adjust the "X-Y" recorder for zero level and perform the test, as in 4.6.8.1, by advancing the frequency pattern through the three frequencies horizontally and then vertically as in 4.6.8.1. Calculate the contrast transfer using the formula $\frac{W - B}{W + B}$ from measured data. Compute the required contrast transfer

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using the values in 3.12.4 and the values of U_T. Contrast transfer on-axis shall conform to the requirements of 3.12.4. Special care must be taken before test data is made to avoid the possible introduction of contrast transfer losses resulting from the test instrument.

4.6.10.8 Luminance gain. - Perform this test using a 2870° ± 100°K tungsten light source, a diffuser with mask, collimator, an eye corrected photo-detector with a 7.0 to 9.5mm entrance pupil, and a calibrated foot lambert meter set up and oriented as in Figure 10. The calibrated foot lambert meter shall be used to measure luminance gain by comparing the luminance seen by the periscope with the luminance at the periscope eye-piece. The light source collimator shall be calibrated to present to the periscope a uniform target luminance of 1.5 x 10^{-3} foot lamberts, with illumination from a 2870° ± 100°K tungsten lamp, subtending a solid angle of 4.5 degrees of periscope field of view centered on the optical axis. Energize the periscope and measure the luminance at the eyepiece with an eye corrected brightness meter which has been calibrated for light from 2870° ± 100°K tungsten lamp and has an entrance pupil of between 7.0mm and 9.5mm. The gain is the rationof the luminance at the eyepiece divided by the target luminance. The luminance gain shall be not less than the value specified in 3.12.5.

4.6.10.9 Stray light luminance. - Perform this test using a calibrated foot lambert meter, an eye corrected photo-detector, a 2870° ± 100°K tungsten light source, a surface of white surround conforming to 3.12.6, and a disk with a black surface luminance of less than 1 x 10-4 foot lamberts set up and oriented as in Figure 13. The instrument described above and in Figure 9 shall be used to measure the stray light luminance. The tungsten light source shall be so positioned as to provide a nearly uniform luminance of 5 times 10^{-3} foot lamberts over the surface of the white surround. The eye corrected photo-detector shall have an aperture of 7.0mm to 9.5mm. Adjust the periscope for infinity focus. Compare the eyepiece luminance with light source off and light source on. Any change in eyepiece luminance, in the measurements with the light source off and the light source on, shall not exceed the tolerance specified in 3.12.6.

4.6.10.10 Field of view. - Perform this test with the periscope positioned on a holding device which permits the periscope to be rotated in a horizontal plane. Project a crossline test pattern onto the objective lens Of the periscope by means of a diffuse 2870° A 100°K tungsten lamp. The periscope shall then be pivoted either to the left or right until the center point of the test pattern seen through the periscope is at the edge of the field of view. the periscope shall then be pivoted in the opposite direction until. the center point of the cross is at the opposite edge of the field of view. The total angle through which the periscope was pivoted shall be measured. The field of view of the periscope shall be not less than the value specified in 3.12.7.

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4.6.10.11 Definition and visibility of projected reticle pattern, clear, medium and dark filters. - Cover the entrance window of the head assembly and project the reticle pattern into the periscope optical system. The image of the reticle pattern shall conform to the requirements of 3.12.8.1 when the filter is rotated to the clear and medium positions. Rotate the filter to the dark position and remove the cover from the entrance window. The projected reticle pattern shall again conform to the requirements of 3.12.8.1 when diffuse illuminance from a 2870° ± 100°K tungsten lamp is presented to the entrance window of the head assembly. The diffuse illuminance shall conform to the requirements of 3.12.8.1b.

4.6.10.12 Reticle accuracy of projected reticle pattern. - The angular value between the boresight cross and maximum range graduation of the projected reticle pattern shall be tested utilizing a collimator reticle target. The target shall be projected by means of a diffuse 2870° ± 100°K tungsten lamp. The accuracy of the projected reticle pattern shall meet the requirements of 3.12.8.2.

4.6.11 Accuracy. - The following tests shall be conducted with the periscope positioned as specified in 3.12.9 on special testing equipment conforming to 4.5.2.3.2. The reticle pattern of the target collimator shall be projected by means of diffuse luminance from a 2870° ± 100°K tungsten lamp. Luminance shall not exceed 1 x 10^{-1} foot lamberts. Unless otherwise specified the tests for accuracies of the periscope shall be performed with the target image adjusted for infinity focus.

4.6.11.1 Flicker - image steadiness. - Project the reticle pattern into the optical train. Position an auxiliary telescope at the eyepiece of the periscope. The auxiliary telescope shall contain a crossline reticle and be capable of focusing on the screen of the image tube. Observe through the auxiliary telescope and superimpose the horizontal and vertical reticle lines on the target image lines on the screen of the image tube. Observe the target image for a time period of not less than one minute. During this one minute time period there shall be no visible flicker or movement of the image as specified in 3.12.9.1.1. Image movement shall be determined by means of the target image with respect to the auxiliary telescope reticle pattern.

4.6.11.2 Sparking - image steadiness. - Perform this test subsequent to, and utilizing the procedure specified in 4.6.11.1. Observe the screen of the energized image tube for a time period of not less than one minute. During the one minute period of observation there shall be no visible or audio evidence of sparking or arcing, as specified in 3.12.9.1.2.

4.6.11.3 Collimation - elevation and deflection. - This test, and the tests specified in 4.6.11.10 through 4.6.11.14 shall be performed with the body assembly removed from the mounting plate assembly. Position the

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calibrated penta prism in the reflected light path of the head assembly so that its incident surface lays in a plane that is parallel to Datum Plane “A”. The emergent surface of the prism shall lay in a plane that is parallel to Datum Plane “D”. Position the viewing telescope in line with the prism so that its optical axis is maintained parallel to Datum Plane “A”, and perpendicular to Datum Plane "D". The horizontal and vertical target image for this test shall be projected in white light. The horizontal and vertical reticle lines of the viewing telescope shall be superimposed on the target image lines prior to positioning the head assembly in the line of sight. Position and secure the head assembly in the holding device. Rotate the input coupling to the angular value specified in 3.12.9.2. Observe the target image through the viewing telescope. Any adjustment of the viewing telescope micrometer knobs to obtain coincidence of the collimator reticle lines and the target image lines is the amount of collimation error. Collimation of the head asseumbly shall meet the requirements of 3.12.9.2.1 and 3.12.9.2.2.

NOTE: Caution shall be exercised during all tests of the head assembly to prevent the input coupling from striking hard againat the travel stops.

4.6.11.4 Image tilt - unity-power. - Position the input coupling as in 4.6.11.3. Position an auxiliary telescope containing a crossline reticle pattern at the exit aperture of the unity-power system. Plumb the vertical reticle line of the telescope and superimpose the reticle line on the vertical target line image. The image of the vertical target line shall be parallel to the telescope vertical reticle line within the tolerance specified in 3.12.9.3.1, as measured on the target.

4.6.11.5 Image tilt - nine-power system. - Perform this test as in 4.6.11.4 utilizing the viewing collimator positioned at the periscope eyepiece. Image tilt of the nine-power system shall conform to the requirements fo 3.12.9.3.2.

4.6.11.6 Parallelism of reticle and image. - Project the reticle pattern into the nine-power system. Remove the viewing telescope from the line of sight and position the input coupling as specified in 3.12.9d. Perform the test when one extremity of the indicated projected reticle line is coincident to the horizontal target line image. The reticle line shall be parallel to the target line within the tolerance specified in 3.12.9.4, as measured on the target.

4.6.11.7 Excursion range, boresight movement. - Establish coincidence of the reticle boresight point with the boresight mark of the projected target image by means of the aximuth end elevation boresight knobs. Rotate the azimuth knob to its clockwise stop, and then rotate the elevation knob to its clockwise stop, then to its counterclockwise stop. Re-establish coincidence of the reticle boresight point with the boresight mark of the target image by means of the boresight knobs. Rotate the azimuth knob to

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its counterclockwise stop, and then rotate the elevation knob to its clockwise stop, then to its counterclockwise stop. These rotations of the azimuth and elevation knobs shall move the reticle boresight point the distance specified in 3.12.9.5.1, as measured on the target.

4.6.11.8 Reticle shift - azimuth and elevation boresight movement. - Establish coincidence of a horizontal reticle line with the horizontal target line image. Rotate the boresight knobs in each direction through their full excursion. During the rotation of the knobs the reticle boresight mark shall not deviate in a direction perpendicular to the intended movement more than the tolerance specified in 3.12.4.5.2.

4.6.11.9 Backlash - azimuth and elevation boresight movement. - With backlash excluded for the direction of travel, establish coincidence of the reticle boresight point with the boresight mark of the projected target image. Record the position of the boresight knobs. Rotate the knobs to their clockwise stops, then with backlash included, re-establish coincidence of the boresight point and boresight mark without overtravel. Record the position of the boresight knobs. Any difference between the two recorded readings is the amount of backlash. Backlash shall not exceed the tolerance specified in 3.12.9.5.3 as measured on the knob scales. Should overtravel occur during the rotation of the boresight knobs, the test is nullified. When overtravel occurs the test for backlash shall be repeated.

4.6.11.10 Elevation accuracy. - Perform this test in sequence to the test in 4.6.11.3. The auto-collimating device specified in 4.5.2.3.2.1 in conjunction with the penta prism and viewing telescope shall be utilized in this test. The mirror shall be auto-collimated to the viewing telescope when Surface "C" of the input coupling of the head assembly is positioned 67 degrees 30 minutes with respect to Datum Plane “A”. The reflecting (auto-collimating) surface of the mirror shall lay in a plane parallel to Datum Plane "D" and a vertical center line passing through the geometric axis of the shaft of the input coupling adapter. The auto-collimating surface of the mirror shall also remain parallel to the vertical ceriter of the input coupling adapter during rotation through its full excursion of travel. Auto-collimation of the mirror and the viewing telescope shall be accomplished with the head assembly removed from the line of sight. Position and secure the head assembly in the holding device with the auto-collimating device affixed and oriented on the input coupling of the head assembly. With backlash removed for travel in elevation, position Surface “C” of the input coupling so that the line of sight is deflected 18 degrees below the horizontal plane. Direct a line of sight through the viewing telescope and note the position of its reticle lines with respect to the reflected reticle lines. While observing the intersecting point (boresight point) of the vertical and horizontal reticle lines of the real and reflected image rotate Surface “C” of the input coupling, with respect to Datum Plane “A”, so that the line of sight is deflected from 18 degrees below the horizontal plane to 22 degrees

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above the horizontal plane. During this rotation of the input coupling the reflected reticle image shall not deviate from the real reticle image beyond the tolerance specified in 3.12.9.6.4 as measured on the viewing telescope.

4.6.11.11 Input coupling - excursion range. - Perform this test in conjunction with the test in 4.6.11.10. Rotation of the input coupling shall conform to the requirements of 3.12.9.6.1.

4.6.11.12 Plumb level. - Perform this test in sequence to the test in 4.6.11.10. Position Surface "C" of the input coupling at 67 degrees 30 minutes with respect to Datum Plane "A". While observing the boresight point of the real and reflected image rotate Surface "C" of the input coupling the maximum elevation, then to maximum depression. During this rotation, the reflected image shall not deviate in azimuth from the real image beyond the tolerance specified in 3.12.9.6.3.

4.6.11.13 Backlash - input coupling. - Perform this test in sequence with 4.6.11.12. With backlash removed for direction of travel, position Surface "C" of the input coupling at 67 degrees 30 minutes with respect to Datum Plane "A". Record the position of the boresight point of the reflected image, as measured on the viewing telescope. Rotate Surface "C" of the input coupling approximately 10° in the same direction of travel, then reverse rotation of the input coupling (with backlash included) until Surface "C" of the input coupling is repositioned 67 degrees 30 minutes with respect to Datum Plane "A", without overtravel. Record this position of the boresight point of the reflected image, as measured on the viewing telescope. Any difference between the recorded readings is the amount of backlash. Backlash shall not exceed the limit specified in 3.12.9.6.5.

NOTE: During this operation (4.6.11.13) if overtravel occurs, the test shall be nullified and procedure specified for backlash shall again be performed.

4.6.11.14 Input coupling travel. - Rotation of the input coupling through its full excursion of travel shall be subjected to a tactile examination while observing the input coupling and the travel stop. Coupling travel shall conform to the requirements of 3.12.9.6.2.

NOTE: Upon completion of the tests in 4.6.11.3 and 4.6.11.10 through 4.6.11.14, the body assembly shall be reassembled to the mount assembly for the remaining periscope tests.

4.6.11.15 Zero setting of diopter scale. - A calibrated dioptometer shall be utilized for measuring the accuracy of zero diopter setting of the diopter knob. Set the dioptometer diopter scale to zero, and adjust the dioptometer eyepiece for best focus of the dioptometer reticle. Set

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the diopter knob of the body assembly to indicate zero diopter. Position the dioptometer at the eyepiece. While viewing through the dioptometer, obtain best focus of the phosphor grain on the screen of the image tube by adjustment of the diopter knob. The phosphor grain on the screen of the image tube shall be in best focus within the tolerance specified in 3.12.9.7.1, as measured on the scale of the diopter knob. Rotation of the diopter knob to measure the accuracy of the zero setting of the diopter scale shall be made with backlash removed for direction of travel.

4.6.11.16 Excursion range - diopter movement. - Perform this test utilizing the dioptometer specified in 4.6.11.15. Position the periscope diopter knob to the zero setting established in 4.6.10.4.15, with backlash removed for direction of travel. Set the dioptometer diopter scale to -3.5 diopters and add or subtract, as applicable, the zero setting value obtained in 4.6.11.15 to the -3.5 diopter, e.g.; should the zero diopter setting of the knob be -0.25 the dioptometer diopter scale shall be set to -3.75 diopters; should the zero diopter setting of the knob be +0.125 the dioptometer diopter scale shall be set to -3.375 diopters. With the dioptometer diopter scale set to the calculated minus diopter value, position the dioptometer at the eyepiece Of the periscope. While observing the screen of the image tube, through the dioptometer, slowly rotate the periscope diopter knob to its minus rotational stop. At one position of the knob, during this rotation of the diopter knob through the minus diopter adjustment range, the phospor grain of the image tube shall be in best focus. Reposition the diopter knob to its established zero setting with backlash removed for direction of travel. Set the dioptometer diopter scale to the calculated plus diopter value, and position the dioptometer at the eyepiece of the periscope. Measure the plus diopter movement excursion range be using the above procedures. The excursion range of the diopter movement shall conform to the requirements of 3.12.9.7.2.

4.6.11.17 Backlash - diopter movement. - Perform this test utilizing the dioptometer specified in 4.6.11.15. The dioptometer shall be zeroed and adjusted as specified therein. Starting from the stop position, rotate the periscope diopter knob clockwise to the “O” position without reversing direction of rotation. Position the dioptometer at the periscope eyepiece and obtain best focus of the target image by adjustment of the dioptometer diopter scale. Record the dioptometer scale reading. Rotate the periscope diopter knob clockwise to its stop position, then return it counterclockwise to the “O” position without overtravel. Again obtain best focus of the target image by adjustment of the dioptometer diopter scale. The difference between the first and second dioptometer diopter scale readings shall not exceed the value specified in 3.12.9.7.3.

4.6.11.18 Focus range. - The test for short range focus requirements of the periscope shall be performed with the target image adjusted to a focus of 40 feet. With the diopter knob positioned for best focus of the phosphor grain on the screen of the image tube, rotate the focus knob of the periscope to the short range setting. The periscope shall conform to the requirements specified in 3.12.9.8.1. The test for infinity (long

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range) requirements of the periscope shall be performed with the target image adjusted for infinity focus. Rotate the focusing knob to the infinity setting, the periscope shall conform to the requirements specified in 3.12.9.8.1. The periscope shall then be subjected to the test specified in 4.6.11.19.

4.6.11.19 Image shift due to rame focus knob. - Adjust the target image for a focus of 1500 meters. With the diopter knob positioned for best focus of the phosphor grain On the screen of the image tube, rotate the focus knob of the periscope and obtain best focus of the target image. Remove the projected reticle pattern of the periscope from the field of view. Position an auxiliary telescope that contains a crossline reticle at the eyepiece of the periscope. Adjust the telescope to obtain coincidence of horizontal and vertical lines of the auxiliary telescope and target image. While observing the target image through the auxiliary telescope, rotate the range focus knob through the clockwise and counterclockwise revolution value specified in 3.12.9.8.2. During this rotation of the focus knob the target image shall not shift in excess of the tolerance specified in 3.12.9.8.2. Image shift shall be measured on the target with reference to horizontal end vertical reticle lines of the auxiliary telescope.

4.6.11.20 Focus - effect on dipter setting. - Adjust the periscope diopter knob for best focus of the phosphor grain on the screen of the image tube. Record the diopter scale reading. Rotate the focusing knob through the range specified in 3.12.9.8.3. Any adjustment of the diopter knob to obtain best focus of the target image, due to rotation of the focusing knob, shall not exceed the tolerance specified in 3.12.9.8.3 as measured on the diopter scale.

4.6.11.21 Image centering. - Position Surface "C" of the input coupling 66 degrees with respect to Datum Plane "A". Projeet a horizontal and vertical crossline image into the nine power system of the periscope. The target image lines shall extend beyond the applicable edge of the field of view of the periscope under test. The target image lines shall also contain graduations in increments of 5.0 mil. Remove the projected reticle pattern of the periscope from the field of via. Observe the target image through the eyepiece of the periscope. The target image shall appear centered in the field of view of the periscope, such that any difference between the number of mils, on the target images measured at the edge of the field does not exceed twice the mil value specified in 3.12.9.8.4 in both the horizontal and vertical direction.

4.6.12 Operability - disengagement of knob locks. - The direct force required to disengage the diopter knob and each boresight knob shall be measured with standard measuring equipment, The measured force shall be not less than or exceed the limits specified in 3.13.1 for the temperature indicated.

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4.6.13 Operability - torque. - The torque required to rotate the input coupling, boresight knobs, diopter knob, and focusing knob shall be measured with standard measuring equipment affixed to the coupling and knobs. The measured torque shall be within the values specified in 3.13.2 at the temperatures indicated. The boresight slip scales and filter lever shall be subjected to a tactile examination and shall conform to the requirements specified in 3.13.2 at the temperatures indicated.

4.6.14 Sealing - assembled periscope. - Assemble the periscope as specified in 3.14a. The internal pressurizing of the assembled periscope to test sealing shall be performed with special testing equipment conforming to 4.5.2.3.4. Secure the poppet in Valve 10516717 in a closed position. Then pressurize the periscope with clean dry air, to the value specified in 3.14a, in accordance with the procedure for internal pressure testing on Sheet 1 of Drawing F8565556. Retain the applied pressure in the periscope for a minimum of 45 minutes. Within the first 30 minutes of this 45 minute time period, the boresight knobs, diopter knob, focus knob, and input coupling shall be rotated through a minimum of 3 full excursions of their rotational range. At the end of the 45 minute period there shall be no evidence of leakage, as indicated on the pressure gauge of the testing equipment, to determine compliance with 3.14a. Upon completion of test release the poppet in the Valve 10516717.

4.6.15 Sealing - head and body assemblies with shipping cover installed. - Performing this test in sequence to the test in 4.6.10.8, utilizing special testing equipment conforming to 4.5.2.3.5. Install the shipping cover on the head assembly and secure the poppet in Valve 10516717 in a closed position. Pressure shall be applied to the head assembly through the valve provided in the shipping cover. Install the shipping cover on the body assembly. Pressureize the head and body asemblies with dry nitrogen gas to the value specified in 3.14b, in accordance with the procedure for internal pressure testing on Sheet 1 of Drawing F8565556. The dry nitrogen gas shall have a dew point at least as low as minus 25°F. Retain the applied pressure in the head and body assemblies for a minimum of 45 minutes. At the end of the 45 minute time period there shall be no evidence of leakage as indicated on the pressure gauge of the testing equipment. Upon completion of this test release the poppet in Valve 10516717.

4.6.16 Flushing. - The special testing equipment specified in 4.6.15 shall be utilized in this test. A dew point tester, with appropriate adapters, shall be used to test the dew point of the interior atmosphere of the head assembly and body assembly. Upon successful completion of the sealing test of the periscope, the head assembly and body assembly shall be removed from its mount. Install the shipping covers on the head assembly and body assembly. The head assembly and body assembly shall then be flushed with dry nitrogen gas. The flushing shall continue until the nitrogen gas bled from the head assembly and body assembly has a dew point at least as low as minus 25°F as measured with the dew point tester. The head assembly and body assembly shall then be sealed. The internal pressure of dry nitrogen gas in the head assembly and body assembly shall be raised to a measured 3 psi ± 0.5 psi. The internal pressure of the head and body assemblies shall be monitored for a period of 5 minutes ± 15 seconds. At the end Of the 5 minutes period there shall

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be no evidence of leakage in the head assembly or body assembly as indicated on the pressure gauge of the testing equipment. The head assembly and body assemblies shall then be subjected to the applicable test specified in 4.6.17 and 4.6.18.

4.6.17 Cleanliness-head assembly. - Perform this test in accordance with the requirements of 3.12.6.1. The accepted head assembly, with shipping covers-installed, shall conform to the cleanliness requirements of 3.16.1.

4.6.18 Cleanliness, body assembly-nine power system. - Energize the image tube and reticle pattern of the body assembly and obtain best focus of the image screen and reticle pattern. While viewing through the eyepiece for evidence of dirt or foreign matter rotate the diopter knob to each stop (cw and ccw), when rotate the focus knob to each stop (cw and ccw). The nine power optical system of the body assembly shall meet the cleanliness requirements of 3.16.2. Any dirt particle that interferes with the legibility of the reticle pattern shall be cause for rejection of the body assembly.

4.6.19 Interchangeability.

4.6.19.1 Interchangeability - mechanical. - this test shall be performed utilizing the maximum and minimum interchangeability gages conforming to Drawings F11738178 and F11738179. The periscopes (head and body assembly) shall be capable of insertion into the above gages. After insertion, the periscopes shall be capable of being firmly secured and subsequently removed.

4.6.19.2 Interchangeability - pressure retention. - This test shall be performed utilizing the mount, pressure test fixture conforming to Drawing F11738184, This pressure retention test shall be conducted utilizing clean dry air. For purposes of this test "clean dry air" is that quality of air that will not deposit dirt or moisture on the components of the assembly under test, when the test is conducted at a temperature between +60° and +90°F. The periscope (head and body assembly) shall be capable of insertion into the above fixture and secured. After insertion the periscope shall be subjected to the sealing test specified in 4.6.14 for a minimum duration of two hours. The assembly shall meet the requirements of 3.17.2.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing and marking shall be in accordance with the applicable packaging data sheet (see 2.1). The level of protection shall be as specified in the procurement document.

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6. NOTES

6.1 Intended use. - The Tank Periscopes: XM44E1 and XF144E2 are intended for use as the gunner's sighting and observation system in the Armored Reconnaissance Airborne Assault Vehicle:M551.

6.2 Ordering data. - Procurement documents should specify:

a. Title, date and number of this specification.
b. licable packaging data sheet number.
c. Selection of applicable levels of preservation packaging and packing.
d. visions for First Article testing.

Custodian:

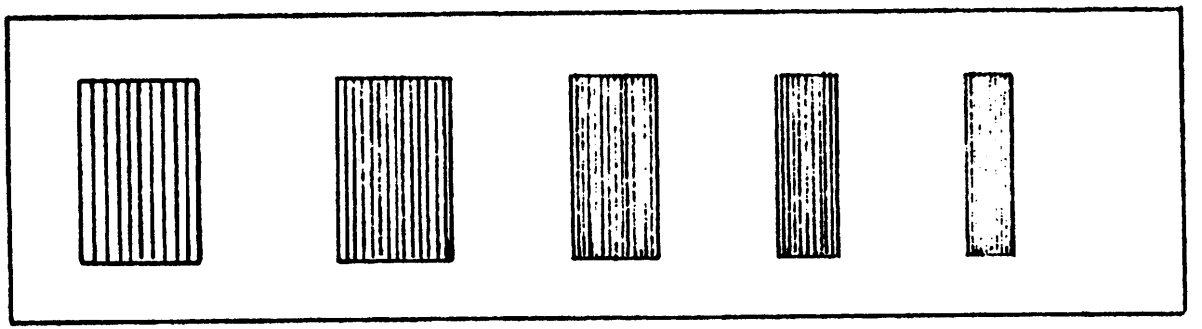
Army - MU

Preparing activity:

Army - MU

Project No. 1240-A245

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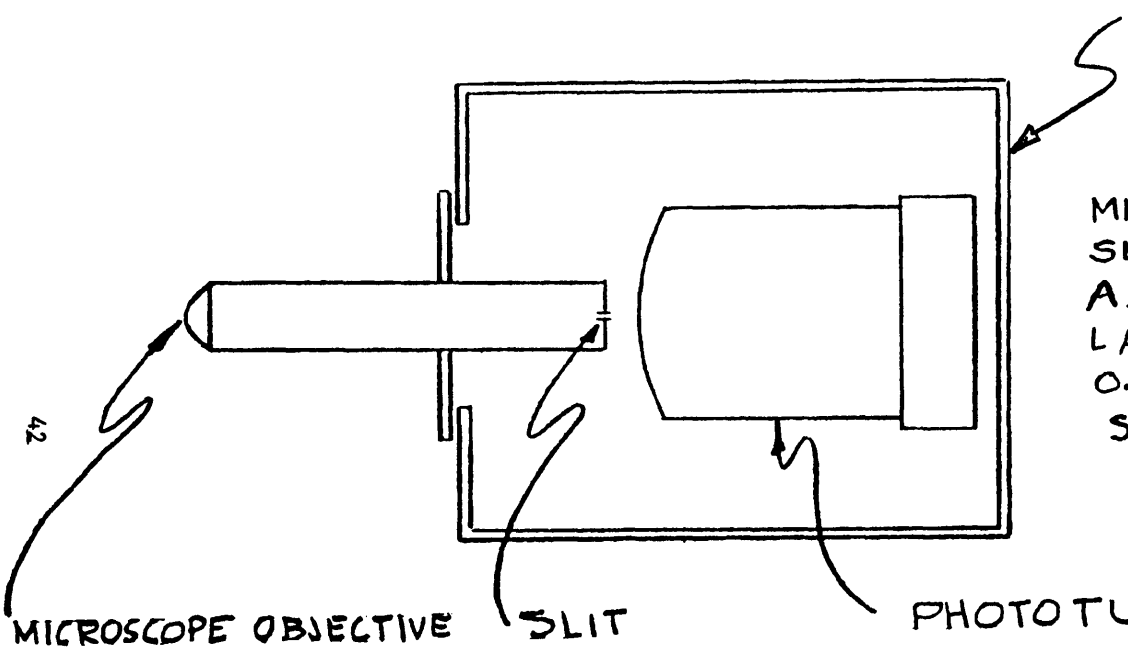
EACH TARGET GROUP WILL HAVE NOT LESS THAN 7 LINE PAIRS OF A PATTERN WHICH WILL GIVE THE INDICATED ANGULAR SUBTENSE

"A": 0.58 LINE PAIR/MILLI-RADIAN - 2.5 L.P./MM AT P.C. OF IMAGE INTENSIFIER.
"B": 1.73 LINE PAIR/MILLI-RADIAN - 7.5 L.P./MM AT P.C. OF IMAGE INTENSIFIER.
"C": 4.62 LINE PAIR/MILLI-RADIAN - 20 L.P./MM AT P.C. OF IMAGE INTENSIFIER.
"D": 6.92 LINE PAIR/MILLI-RADIAN - 30 L.P./MM AT P.C. OF IMAGE INTENSIFIER.
"E": 9.23 LINE PAIR/MILLI-RADIAN - 40 L.P./MM AT P.C. OF IMAGE INTENSIFIER.

FREQUENCY PATTERN - CONTRAST TRANSFER

FIGURE 1

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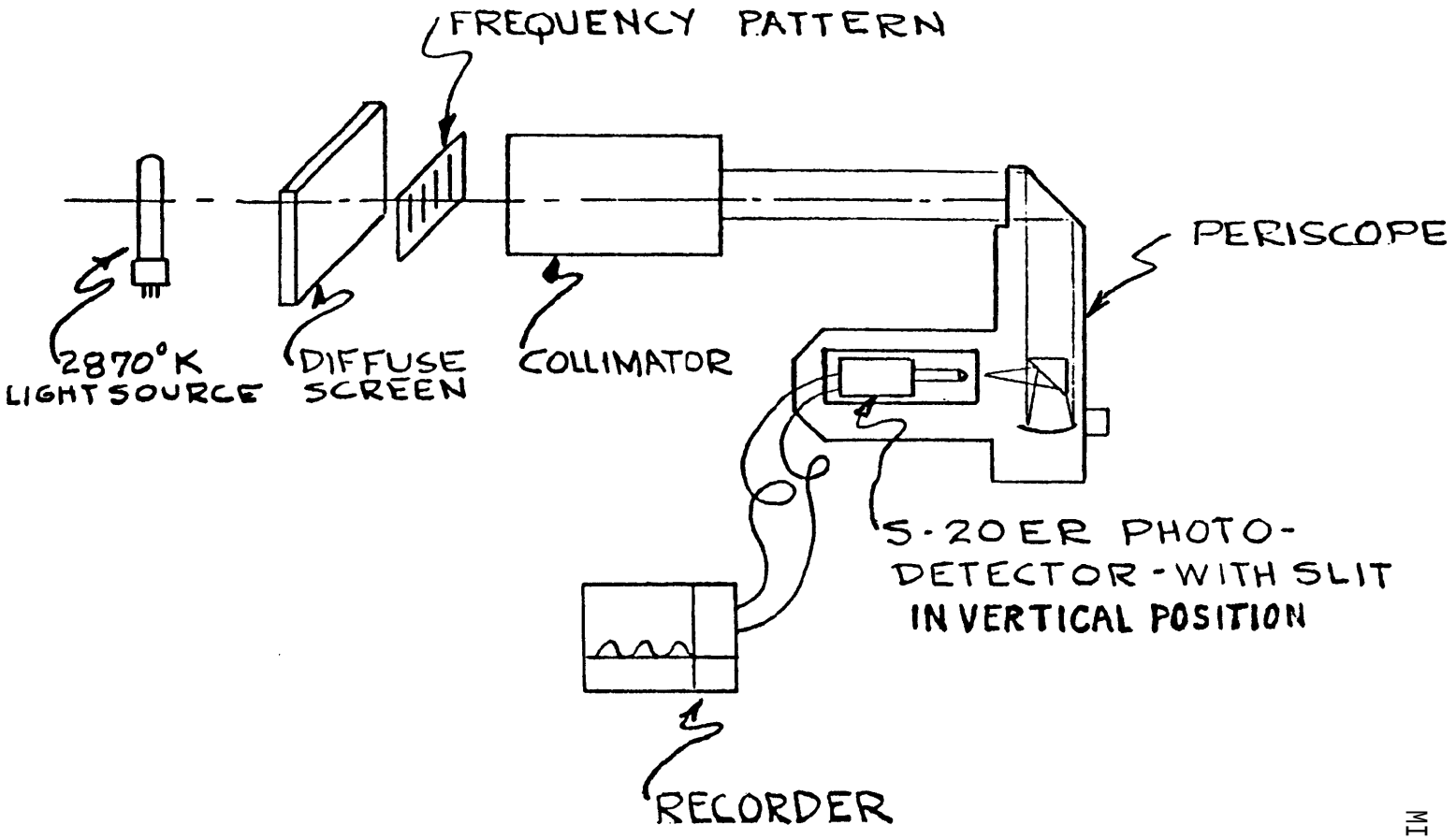


*S-20ER DETECTOR: Standard S-20 Photo-cathod response extended at the red end so that th sensitivity at 0.8 micron and at 0.85 are 85% of the minimum values in the Purchase Description for the Image Intensifier Tube.

S-20ER DETECTOR AND SLIT

FIGURE 2

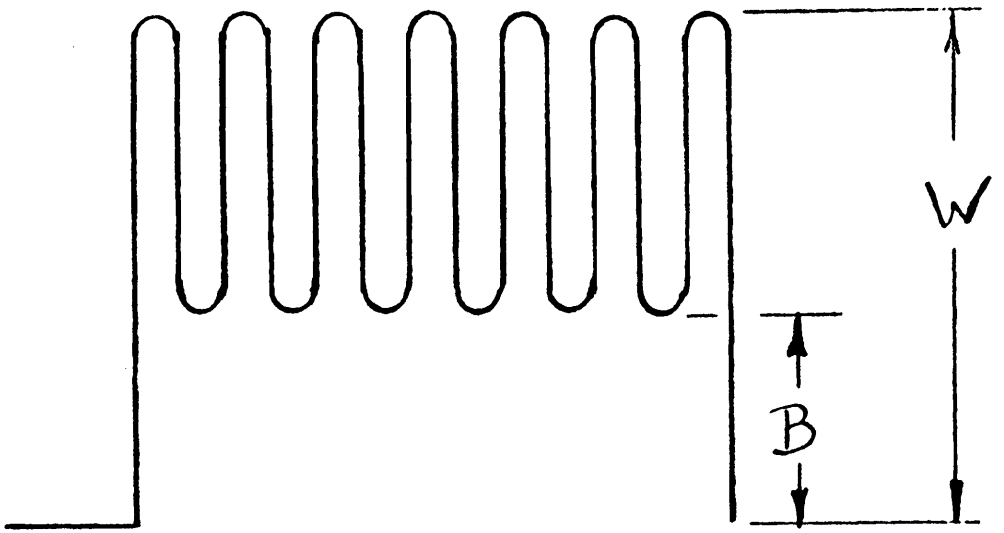
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CONTRAST TRANSFER
PERISCOPE OBJECTIVE

FIGURE 3

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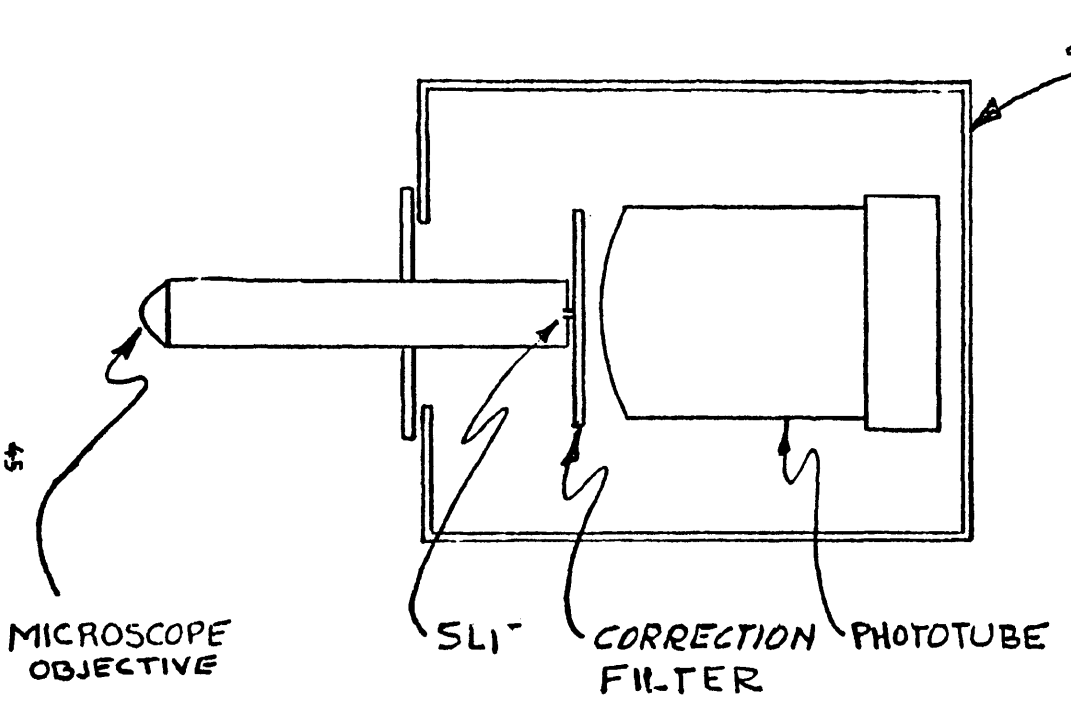


NOTE: CONTRAST TRANSFER IS DEFINED AS $\frac{W-B}{W+B}$

CONTRAST TRANSFER WAVE FORM

FIGURE 4

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MICROSCOPE OBJECTIVE & SLIT ASSEMBLED AS A UNIT. ASSEMBLY MUST MOVE LATERALLY & UP & DOWN 0.3 INCHES (TOTAL 0.6 INCH) SLIT MUST ROTATE 90°.

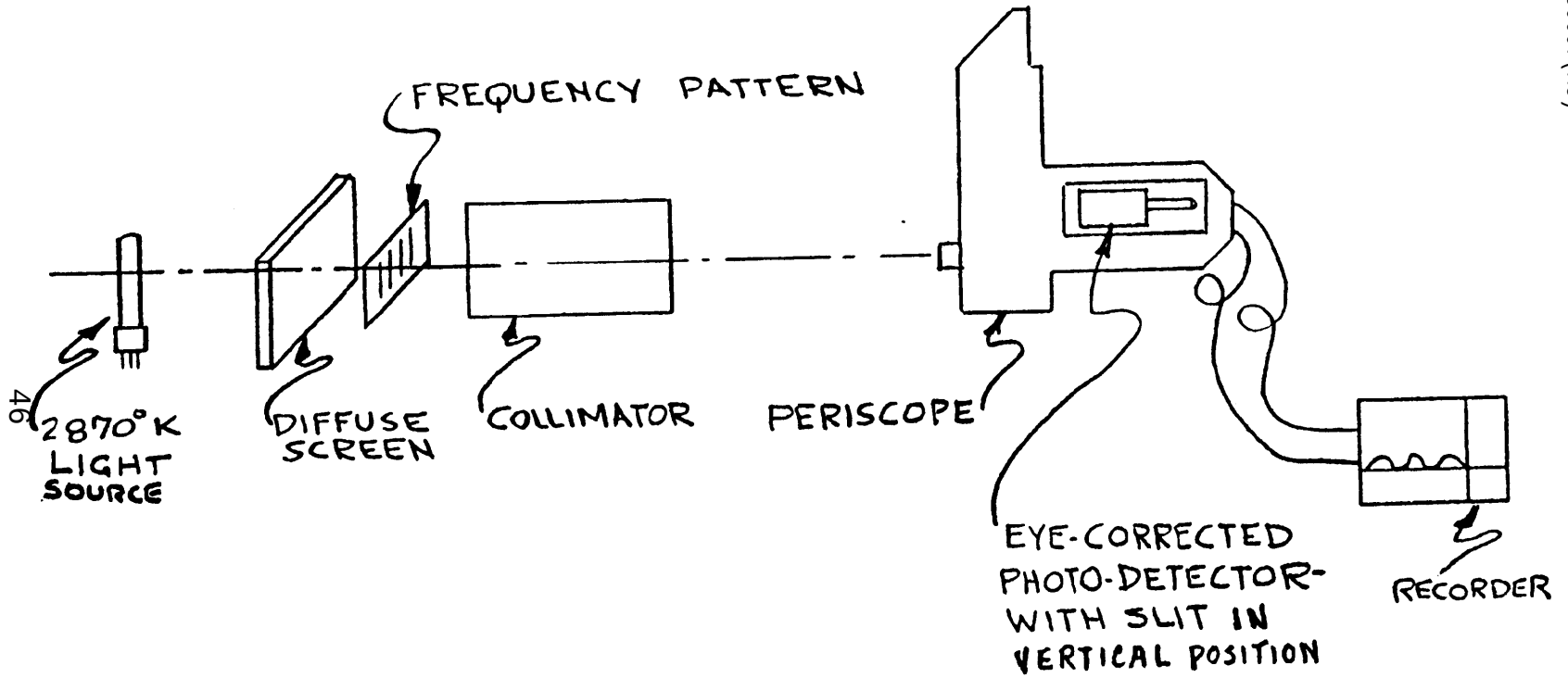
CORRECTION FILTER WILL CHANGE TUBE RESPONSE TO STANDARD VISIBILITY FUNCTION (V_2).

MIL-P-50607(MD)

EYE-CORRECTED DETECTOR AND SLIT

FIGURE 5

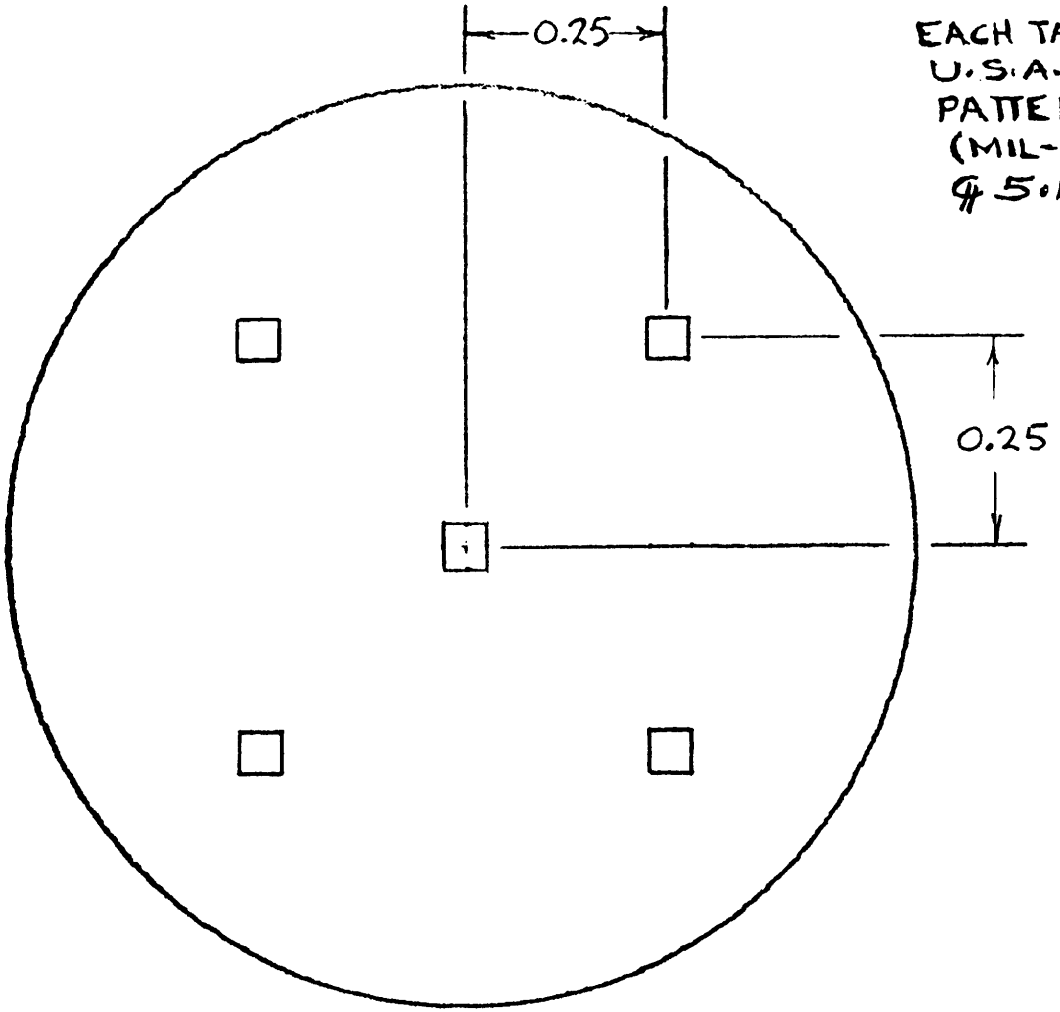
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CONTRAST TRANSFER
PERISCOPE EYEPIECE

FIGURE 6

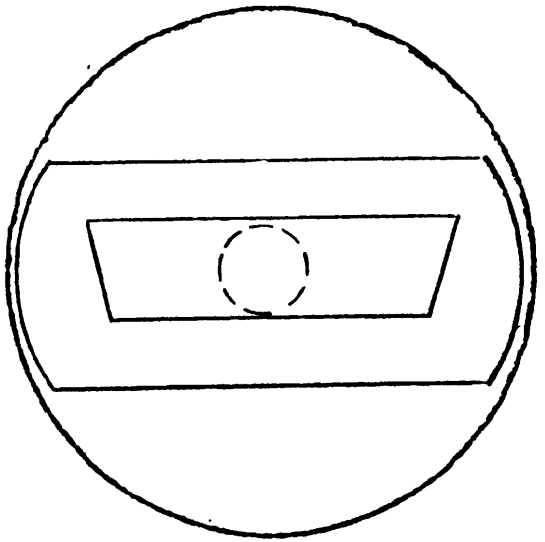
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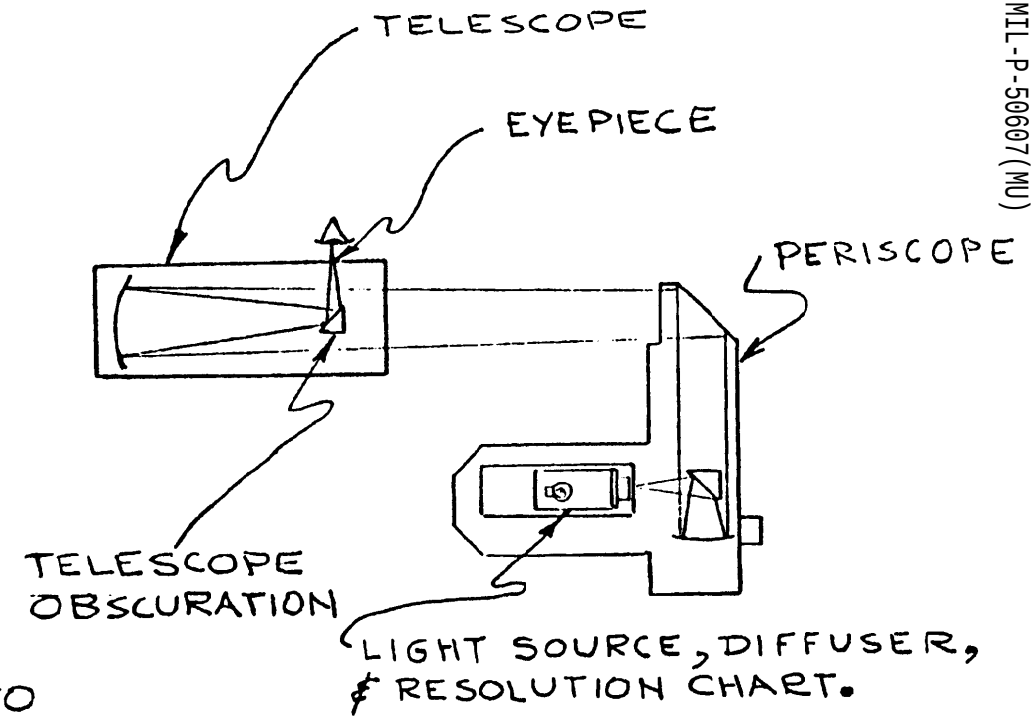


EACH TARGET TO BE A
U.S.A.F. 1951 RESOLUTION
PATTERN UP TO 228 L.P./MM
(MIL-STD-150, 12 MAY 1959)
¶ 5.1.1.7

RETICLE PATTERN
FOR
OBJECTIVE RESOLUTION TEST
FIGURE 7

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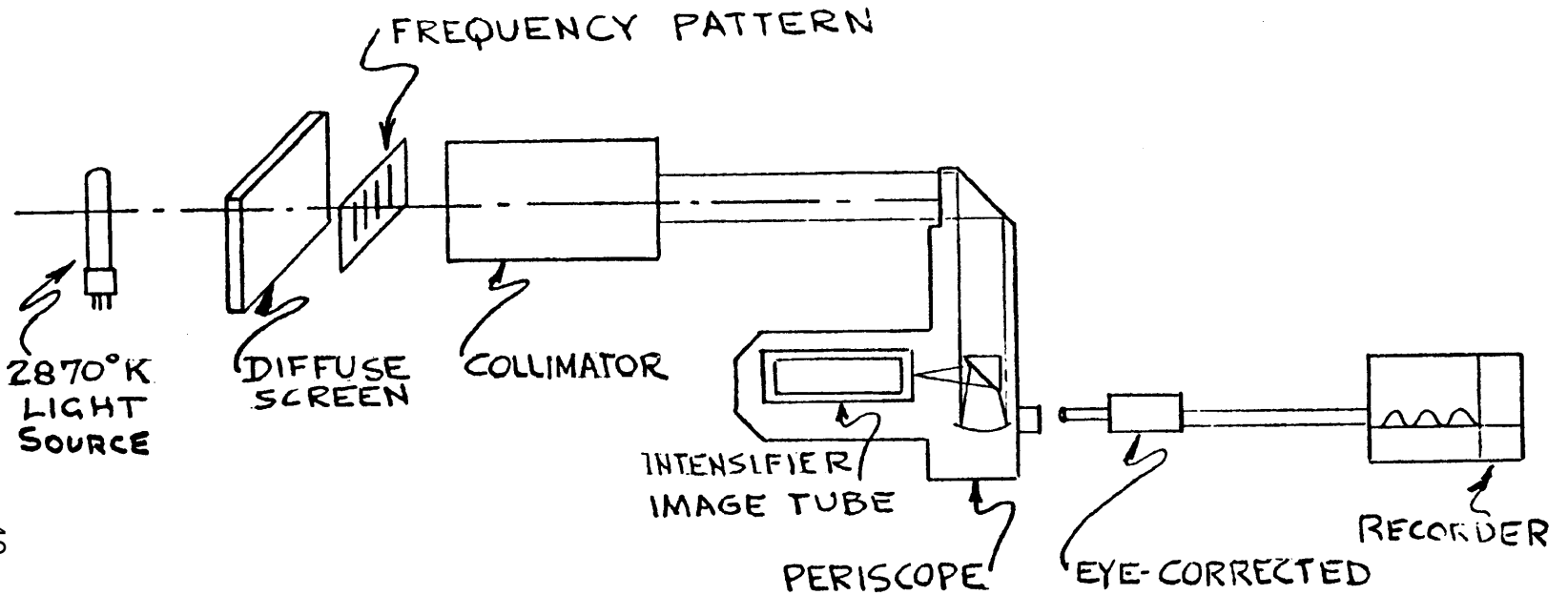


CARE MUST BE TAKEN TO SUPERIMPOSE THE OBSCURATIONS OF THE PERISCOPE AND THE TELESCOPE.

RESOLUTION OBJECTIVE ASSEMBLY

FIGURE 8

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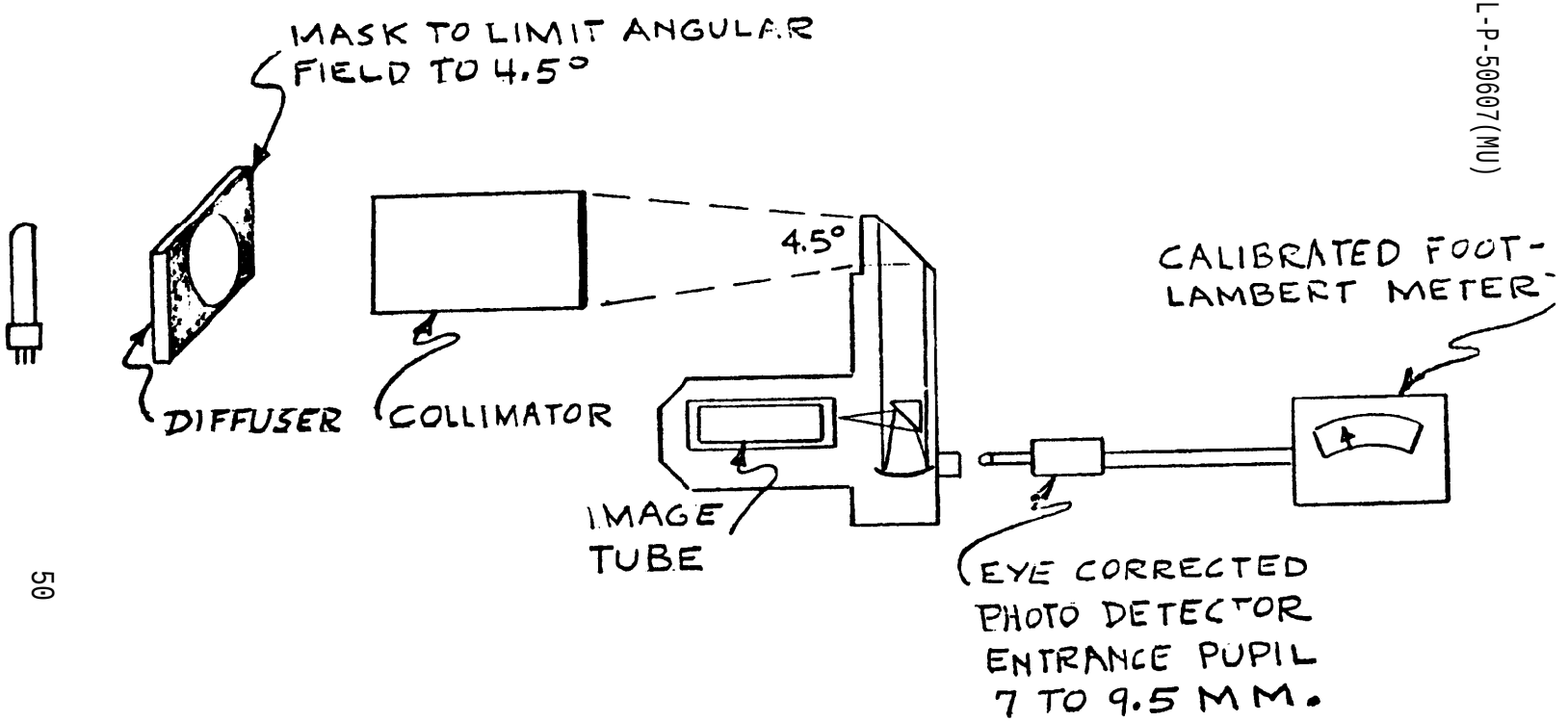
FREQUENCY PATTERN MUST BE CAPABLE OF BOTH HORIZONTAL & VERTICAL OPERATION.

CONTRAST TRANSFER

COMPLETE PERISCOPE

FIGURE 9

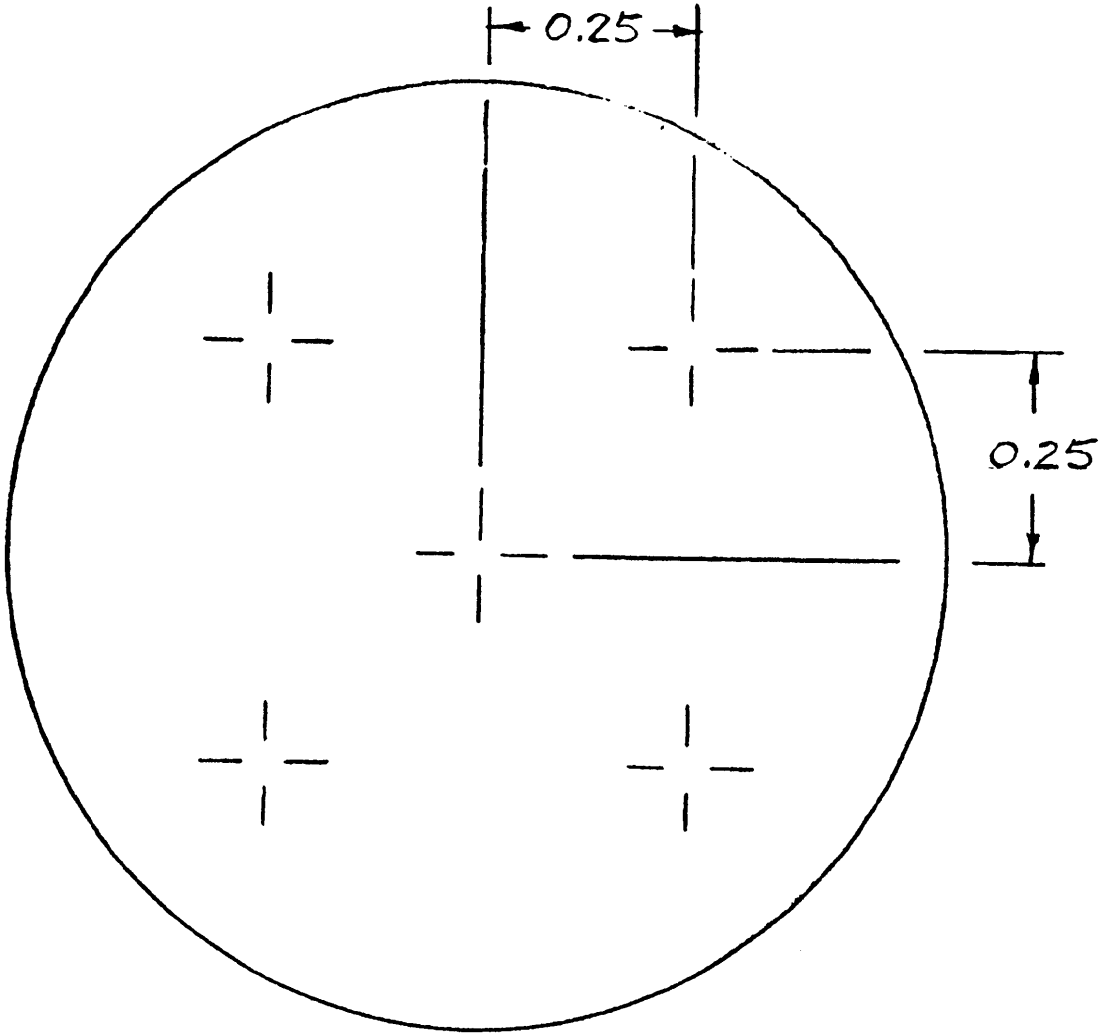
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LUMINANCE GAIN

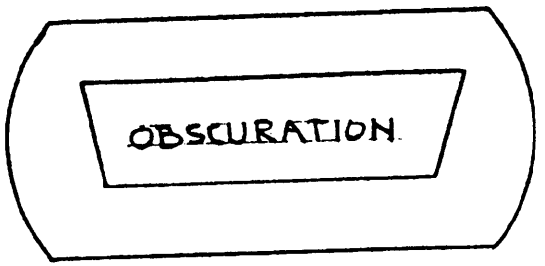
FIGURE 10

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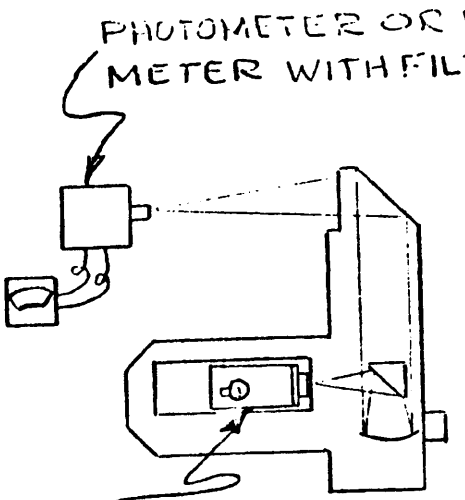


RETICLE PATTERN
FOR
T/# TEST
FIGURE 11

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PERISCOPE ENTRANCE PUPIL AS SEEN FROM IN FRONT OF HEAD.

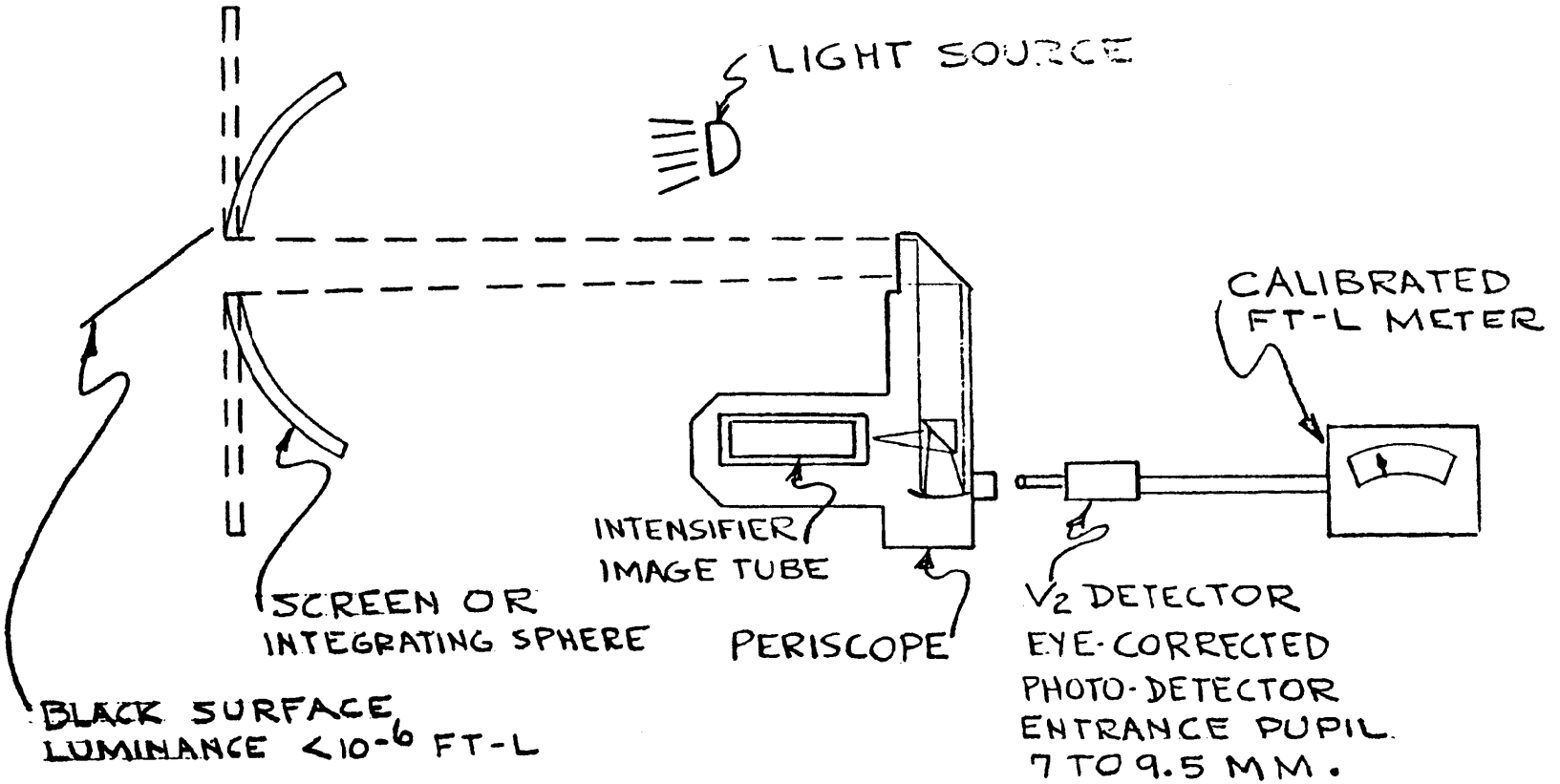


LIGHT SOURCE, DIFFUSER, & RETICLE ASSEMBLY, PLACED IN TUBE CAVITY.

OBJECTIVE TRANSMISSION, VISIBLE & I.R.

FIGURE 12

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STRAY-LIGHT MEASUREMENT

FIGURE 13

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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